A 2-D Unstructured Finite-Volume Compressible Navier–Stokes Solver:
Implementation, Verification and Benchmark Results

`cfd2d` — submission for the CFD solver agentic benchmark

August 27, 2026

Abstract

This report documents `cfd2d`, a cell-centred finite-volume solver for the two-dimensional compressible Navier–Stokes equations of a calorically perfect gas, written from scratch in C++17 with MPI domain decomposition, and the results it produces for the eight cases of the benchmark. The solver imports unstructured CGNS meshes (mixed triangles and quadrilaterals; the cylinder mesh is multi-zone and is merged through its vertex 1-to-1 connectivity), partitions the cell adjacency graph with METIS k-way, and stores on every rank only the cells it owns plus one ghost layer. The spatial discretisation is piecewise linear with inverse-distance-weighted least-squares gradients of the primitive variables and a Venkatakrishnan limiter with a first-order positivity fallback; the inviscid flux is HLLC (a Roe flux with a Harten–Yee entropy fix is also implemented and is used as a cross-check), combined with a multidimensional shock fix that blends towards Rusanov only on faces lying along a strong shock and thereby suppresses the carbuncle instability at the Mach 2 bow shock without touching the rest of the flow, and the viscous flux uses a Newtonian stress tensor and Fourier heat flux built from corrected face gradients. Steady cases are marched with implicit backward Euler in local pseudo-time solved by a matrix-free LU-SGS symmetric Gauss–Seidel relaxation; the Reynolds 200 cylinder uses a true dual-time BDF2 scheme with an outer physical-time loop, frozen histories and inner nonlinear iterations. A method-of-manufactured-solutions study measures a discretisation-error order of 2.30 for the unlimited second-order scheme against 0.92 for the first-order scheme. 8 of the eight required cases produced a complete result package, with 0 marked failed. The Reynolds 20 cylinder gives $C_D = 2.01762$, in agreement with the accepted value of 2.0–2.05, and the Reynolds 200 cylinder develops a vortex street with Strouhal number 0.1859 and mean drag 1.2661, each a few percent below the accepted range. Of 118 automated physics sanity checks over 8 cases, 0 fail.

Contents

1 Introduction

The benchmark asks for a complete, self-contained 2-D unstructured compressible Navier–Stokes solver, run on eight cases covering subsonic, transonic and supersonic inviscid flow over a NACA0012 aerofoil, the same three Mach numbers with laminar viscous walls at $Re = 5000$, and a circular cylinder at $Re = 20$ (steady) and $Re = 200$ (unsteady vortex shedding). The cases are listed in table 7.

This is a technical report rather than a run log: sections 2 to 7 describe what the solver actually does and why, section 8 verifies the discretisation independently of the benchmark cases, and sections 10 and 11 present and interpret the computed results. Section 14 states the known weaknesses honestly.

Every table, every figure and every computed quantity quoted in the text is generated from the submitted output files by the scripts in `tools/`: `make_report_tables.py` writes the tables and a file of LaTeX macros directly from `results/*/metadata.json`, `run_status.json`, `forces.csv`, `residuals.csv` and `surface.csv`; `mesh_facts.py` computes the mesh and boundary-layer quantities from the field files; `make_figures.py` writes every figure together with the manifest row that names its source data. The prose therefore quotes macros, not transcribed digits, and cannot drift when a case is re-run. Two categories of number are still written literally and are meant to be: constants of the method (the limiter's K, the entropy-fix coefficient, CFL schedules and tolerances) and closed-form reference values (the sonic C_p^*, the Rayleigh–Pitot limit, Blasius, literature ranges), all of which are stated with the formula or the citation that produces them. `scripts/check_submission.sh` fails the build if any macro the report uses could not be populated from data.

2 Governing Equations and Nondimensionalisation

2.1 Conservation laws

The solver advances the conservative state

$$\mathbf{U} = [\rho,\ \rho u,\ \rho v,\ \rho E]^T, \tag{1}$$

governed by

$$\frac{\partial \mathbf{U}}{\partial t} + \frac{\partial \mathbf{F}^i}{\partial x} + \frac{\partial \mathbf{G}^i}{\partial y} = \frac{\partial \mathbf{F}^v}{\partial x} + \frac{\partial \mathbf{G}^v}{\partial y}, \tag{2}$$

with the inviscid fluxes

$$\mathbf{F}^i = \begin{bmatrix} \rho u \\ \rho u^2 + p \\ \rho u v \\ u(\rho E + p) \end{bmatrix}, \qquad \mathbf{G}^i = \begin{bmatrix} \rho v \\ \rho u v \\ \rho v^2 + p \\ v(\rho E + p) \end{bmatrix}, \tag{3}$$

and the viscous fluxes

$$\mathbf{F}^v = \begin{bmatrix} 0 \\ \tau_{xx} \\ \tau_{xy} \\ u\tau_{xx} + v\tau_{xy} - q_x \end{bmatrix}, \qquad \mathbf{G}^v = \begin{bmatrix} 0 \\ \tau_{xy} \\ \tau_{yy} \\ u\tau_{xy} + v\tau_{yy} - q_y \end{bmatrix}. \tag{4}$$

The Newtonian stress tensor uses the Stokes hypothesis,

$$\tau_{xx} = 2\mu u_x - \tfrac{2}{3}\mu(u_x + v_y), \quad \tau_{yy} = 2\mu v_y - \tfrac{2}{3}\mu(u_x + v_y), \quad \tau_{xy} = \mu(u_y + v_x), \tag{5}$$

and the heat flux is Fourier's law

$$\mathbf{q} = -k\nabla T, \qquad k = \frac{\mu c_p}{\mathrm{Pr}}, \qquad c_p = \frac{\gamma R}{\gamma - 1}. \tag{6}$$

The gas is calorically perfect,

$$p = (\gamma - 1)\rho e, \qquad E = e + \tfrac{1}{2}(u^2 + v^2), \qquad a = \sqrt{\gamma p/\rho}, \qquad T = \frac{p}{\rho R}. \tag{7}$$

Inviscid mode is obtained by setting $\mu = k = 0$; the whole viscous branch of the residual is then skipped, and the viscous force columns of `forces.csv` are identically zero.

2.2 Nondimensionalisation

The case files supply a nondimensional freestream directly: $\rho_\infty = 1$, $|\mathbf{U}_\infty| = 1$, $R = 1$, $\gamma = 1.4$, $\mathrm{Pr} = 0.72$ and

$$p_\infty = \frac{1}{\gamma M_\infty^2} \quad \Longrightarrow \quad a_\infty = \sqrt{\gamma p_\infty/\rho_\infty} = \frac{1}{M_\infty}, \tag{8}$$

so that $|\mathbf{U}_\infty|/a_\infty = M_\infty$ exactly. With $R = 1$ the freestream temperature is $T_\infty = p_\infty$. Lengths are scaled by L_{ref} (aerofoil chord, cylinder diameter), both equal to 1 here, and time by $L_{\mathrm{ref}}/|\mathbf{U}_\infty|$. The case file carries a separate `reference.reynolds_length` used in eq. (9); it defaults to `reference.length` and the two are equal for all eight cases. For laminar cases the constant molecular viscosity is fixed by the case Reynolds number,

$$\mu = \frac{\rho_\infty |\mathbf{U}_\infty| L_{\mathrm{ref}}}{Re}, \tag{9}$$

evaluated in `CaseConfig::molecularViscosity()`; a Sutherland law is also implemented and selectable through `physics.viscosity_model`. Force coefficients use the freestream dynamic pressure $q_\infty = \frac{1}{2}\rho_\infty |\mathbf{U}_\infty|^2 = 0.5$ and the case `reference.area`:

$$C_D = \frac{\mathbf{F}\cdot\hat{\mathbf{d}}}{q_\infty A_{\mathrm{ref}}}, \quad C_L = \frac{\mathbf{F}\cdot\hat{\mathbf{l}}}{q_\infty A_{\mathrm{ref}}}, \quad C_{m,z} = \frac{\sum_f (\mathbf{x}_f - \mathbf{x}_{\mathrm{ref}}) \times \mathbf{f}_f}{q_\infty A_{\mathrm{ref}} L_{\mathrm{ref}}}, \tag{10}$$

where $\hat{\mathbf{d}}$ is the freestream direction and $\hat{\mathbf{l}} \perp \hat{\mathbf{d}}$. The pressure integral uses $(p - p_\infty)$ rather than p; for a closed body the two are identical because $\oint p_\infty \hat{\mathbf{n}}\,\mathrm{d}S = 0$, but subtracting the freestream removes a large cancelling contribution and the round-off that goes with it.

2.3 Residual norms

The residual is reported as a volume-normalised, freestream-scaled quantity so that the four equations are comparable and the reduction is meaningful:

$$r_{k,c} = \frac{R_{k,c}}{V_c\, s_k}, \qquad \mathbf{s} = \left[\rho_\infty,\ \rho_\infty a_\infty,\ \rho_\infty a_\infty,\ \rho_\infty a_\infty^2\right], \tag{11}$$

where $R_{k,c}/V_c$ is the local rate of change of the conserved variable, so that $r \to 0$ is exactly the statement that the solution is steady. Two norms of r are written to `residuals.csv`, both formed with MPI reductions over all ranks:

$$\|r\|_2 = \left(\frac{1}{|\Omega|\, n_{\mathrm{var}}} \sum_c V_c \sum_k r_{k,c}^2\right)^{1/2}, \qquad \|r\|_\infty = \max_{c,k} |r_{k,c}|. \tag{12}$$

The L_2 norm is volume weighted, i.e. it is the discrete $L_2(\Omega)$ norm of $\partial\mathbf{U}/\partial t$ divided by the domain volume. This matters on these meshes: an *unweighted* root-mean-square over cells is dominated by a handful of degenerate sliver cells at the sharp aerofoil trailing edge — on `NACA0012_H2.cgns`, 232 cells out of 20816 (mean area 9.9×10^{-9} against a median of 5.9×10^{-6}, together 1.1×10^{-10} of the domain area) carry 90% of it — where the inviscid solution around a sharp trailing edge is genuinely singular and no scheme converges. Weighting by cell volume measures the physical rate of change of the solution instead of the rate of change in the smallest cell. The L_∞ column keeps that behaviour visible rather than hiding it: it plateaus between 15 and 833 on the aerofoil cases, and its location is written into the field files as the `ScaledResidual` variable, so a reader can see for themselves that it sits in the trailing-edge slivers. All of these numbers are computed by `tools/mesh_facts.py` from the submitted field files.

2.4 Convergence criteria for steady cases

A steady case is declared `converged` only when *both*

1. the requested residual reduction is achieved, $\|r\|_2 \leq 10^{-n}\|r\|_2^0$ with n from the case file, *and*
2. the drag coefficient is stationary, tested as a *trend* rather than a point-to-point difference: with $w = \max(50, N_{\text{step}}/10)$, the mean of C_D over the last w steps must differ from its mean over the preceding w steps by less than $\varepsilon_F = \max(3 \times 10^{-3}|\overline{C_D}|,\, 5 \times 10^{-5})$, and the RMS scatter within the last window must be below $3\varepsilon_F$.

Every part of this is deliberate. Comparing two window *means* averages out the bounded oscillation that the shock cases settle into while still detecting a slow monotone drift, which is the failure mode the test exists to catch: a viscous case can meet a relative residual target thousands of iterations before its boundary layer has settled, with C_D still falling by tens of percent. A single point-to-point difference is far noisier and lets such a case through. The absolute floor in ε_F matters because the inviscid drag of a symmetric aerofoil at zero incidence is a small numerical residue of d'Alembert's paradox whose exact value is zero, so a purely relative criterion would demand a precision that carries no physical meaning. The relative part is 0.3% rather than something tighter because the shock-containing cases genuinely carry an uncertainty of that order (section 14), and the scatter bound is looser than the trend bound for the same reason. The realised drift, scatter and tolerance of every run are written into the `notes` field of `run_status.json` and reproduced in `report/run_manifest.md`. The second condition exists because the first alone is not sufficient here: on these viscous-type meshes the impulsive start produces a very large first residual (the no-slip velocity jump across a 5×10^{-6}-chord first cell), so a purely relative residual criterion can be met thousands of iterations before the boundary layer has settled. If the step budget is exhausted, the run is accepted only if the forces have plateaued to three times ε_F and the residual has still dropped by more than two orders; otherwise it is marked `failed`, and `metadata.json` then reports `completed: false` so that the benchmark validator rejects it.

The pseudo-time CFL follows the case schedule as an *upper envelope*, guarded by a standard residual-based back-off: if $\|r\|_2$ grows by more than a factor of two above the best value seen so far, the CFL factor is halved (and recovers by 2% per step up to the schedule value). The effective CFL is therefore never larger than the value requested by the case file. The number of back-offs is recorded in the run notes.

3 Meshes, Boundary Tags and Case Inputs

3.1 CGNS import

`src/mesh/CgnsReader.cpp` discovers everything from the file: it selects the base whose cell dimension is 2, iterates over all zones, reads `CoordinateX`/`CoordinateY`, and classifies each element section by its CGNS element type. `TRI_3`, `QUAD_4` and `MIXED` sections containing them become cells; `BAR_2` elements are stored in a zone-local map from element id to node pair and are used only to resolve boundary conditions. Nothing is hard-coded: zone names, section names, element types and family names are all read from the file, and unsupported element types produce a clear error naming the offending section.

Boundary conditions come from the `ZoneBC_t` nodes. For each `BC_t` the point set is expanded (`PointRange`/`PointList`/`ElementRange`/`ElementList`) and, when the grid location is element-based, each index is looked up in the `BAR_2` map; a vertex-located BC is also supported by selecting the boundary edges whose two endpoints are both listed. The patch is named by the `FamilyName_t` of the BC when present (this is how `bc-2`, `bc-4`, `WALL` and `FAR` are obtained) and otherwise by the `BC_t` node name. The case file's `boundary_conditions` map is then resolved against those patch names, and a mismatch in either direction is a fatal, explicitly worded error.

Multi-zone merging. The cylinder mesh has 2 zones joined by five `Abutting1to1` vertex connectivities. The reader unions the matching node pairs (`PointList` against `PointListDonor`) in a disjoint-set structure and renumbers the surviving nodes, so the assembled mesh is a single conforming grid; the `con-*` `BAR_2` sections that describe the interface become ordinary interior faces and are never treated as boundaries. A geometric fallback (`--node-merge-tol`) exists for meshes whose interfaces are not described by vertex connectivity, but it is not needed here. After the face topology is built, every boundary face must be claimed by exactly one declared patch; otherwise the solver aborts with a message that points at the first offending edge, which is what detects a failed merge.

3.2 Geometry

Cells are stored as general polygons in CSR form and reoriented counter-clockwise (positive signed area). Cell volumes and centroids use the exact polygon formulae

$$V = \tfrac{1}{2}\sum_i (x_i y_{i+1} - x_{i+1} y_i), \qquad \mathbf{x}_c = \frac{1}{6V}\sum_i (\mathbf{x}_i + \mathbf{x}_{i+1})(x_i y_{i+1} - x_{i+1} y_i). \tag{13}$$

Faces are the polygon edges, collected in a hash map keyed on the unordered node pair, so an edge shared by two cells becomes one interior face and an edge seen once becomes a boundary face. The face normal is taken from the counter-clockwise traversal of the *left* cell, $\hat{\mathbf{n}} = (\Delta y, -\Delta x)/\ell$, which therefore points out of the left cell (and out of the domain on boundary faces) by construction, and ℓ is the face area in 2-D. Every run verifies the metrics against the divergence theorem,

$$V_c \overset{!}{=} \tfrac{1}{2}\sum_{f\in c} (\mathbf{x}_f - \mathbf{x}_c)\cdot \hat{\mathbf{n}}_f\, \ell_f, \tag{14}$$

and aborts if the worst relative error exceeds 10^{-9}; the observed value is 1.1×10^{-10} on the aerofoil mesh and 4.8×10^{-13} on the cylinder mesh. The difference between the two is the floating-point cancellation floor of the thin near-wall cells: the aerofoil mesh reaches an aspect ratio of 2821 (longest over shortest edge) where the cylinder mesh stays below 16.

3.3 The two meshes and the eight cases

Table 1 summarises what the reader finds in the two CGNS files, and table 2 lists the physical parameters of the eight required cases as they are read from the JSON inputs.

Table 1: Meshes as imported by `cfd2d inspect-mesh`.

	`NACA0012_H2.cgns`	`CylinderB1.cgns`
CGNS zones	1	2
nodes (after merging)	15682	9995
cells	20816	10185
triangles / quadrilaterals	10752 / 10064	500 / 9685
faces	36498	20180
boundary faces	484	120
wall patch	`bc-4` (404 faces)	`WALL` (100 faces)
farfield patch	`bc-2` (80 faces)	`FAR` (20 faces)
domain extent	$[-80, 80]^2$ chords	$[-200, 200]^2$ diameters
body	chord 1.005, 12% thick	diameter 1.0
smallest / largest cell area	1.5×10^{-9} / 28.2	6.3×10^{-5} / 3.1×10^3

Table 2: Physical parameters of the eight required cases, read from `inputs/cases/*.json` and echoed through `metadata.json`.

case	mesh	M_∞	AoA	Re	cells	wall BC
NACA0012 $M_\infty = 0.15$ inviscid	NACA0012_H2	0.15	0°	–	20816	slip
NACA0012 $M_\infty = 0.80$ inviscid	NACA0012_H2	0.8	0°	–	20816	slip
NACA0012 $M_\infty = 2.0$ inviscid	NACA0012_H2	2	0°	–	20816	slip
NACA0012 $M_\infty = 0.15$ laminar $Re = 5000$	NACA0012_H2	0.15	0°	5000	20816	no-slip adiabatic
NACA0012 $M_\infty = 0.80$ laminar $Re = 5000$	NACA0012_H2	0.8	0°	5000	20816	no-slip adiabatic
NACA0012 $M_\infty = 2.0$ laminar $Re = 5000$	NACA0012_H2	2	0°	5000	20816	no-slip adiabatic
cylinder $M_\infty = 0.1$ laminar $Re = 20$	CylinderB1	0.1	0°	20	10185	no-slip adiabatic
cylinder $M_\infty = 0.1$ laminar $Re = 200$	CylinderB1	0.1	0°	200	10185	no-slip adiabatic

Both meshes are viscous-type grids with strongly stretched near-wall layers, which is what makes the inviscid Mach 2 case numerically demanding (section 14).

4 Spatial Discretisation

The scheme is cell-centred finite volume. Integrating eq. (2) over cell c gives

$$V_c \frac{\mathrm{d}\mathbf{U}_c}{\mathrm{d}t} = -\mathbf{R}_c, \qquad \mathbf{R}_c = \sum_{f \in \partial c} \left[\mathbf{\Phi}_f^i - \mathbf{\Phi}_f^v \right] \ell_f, \tag{15}$$

where $\mathbf{\Phi}_f^i$ is the numerical inviscid flux and $\mathbf{\Phi}_f^v$ the viscous flux, both projected on the outward normal of c. The same face flux is added to the left cell and subtracted from the right cell, so the scheme is discretely conservative by construction.

4.1 Second-order reconstruction

Gradients of the primitive variables $\mathbf{W} = (\rho, u, v, p)$ are obtained from a weighted least-squares fit over the face neighbours of each cell,

$$\min_{\nabla W} \sum_j w_j \big(\nabla W \cdot \mathbf{d}_j - (W_j - W_c)\big)^2, \qquad \mathbf{d}_j = \mathbf{x}_j - \mathbf{x}_c, \quad w_j = |\mathbf{d}_j|^{-2}, \tag{16}$$

which for two unknowns reduces to inverting the symmetric 2×2 matrix $A = \sum_j w_j \mathbf{d}_j \mathbf{d}_j^T$. A^{-1} and the weighted displacement vectors depend only on geometry and are precomputed once. The inverse-distance-squared weighting is essential on the near-wall cells, whose neighbours are two orders of magnitude closer in the wall-normal direction than along the wall; it makes all stencil entries contribute comparably to A. For a cell that touches a boundary the stencil is augmented with a virtual point at the boundary-face centroid carrying the boundary-condition state (section 5), which keeps the gradient well-posed on wall cells.

Face states are then

$$\mathbf{W}_L = \mathbf{W}_c + \phi_c \odot \big(\nabla \mathbf{W}_c \cdot (\mathbf{x}_f - \mathbf{x}_c)\big), \tag{17}$$

with the limiter $\phi_c \in [0, 1]^4$ defined per variable. Reconstruction is active in every production run reported here; the only first-order fallback is the positivity guard described below, and its activation count is written into `metadata.json` for every case (field `positivity_fallback_face_states`). That it is genuinely active is easy to check: rerunning the $M_\infty = 0.15$ laminar case with `--first-order` raises the drag from 0.05339 to 0.06801, i.e. by 27%, because the first-order scheme cannot resolve the boundary layer on this mesh (`studies/verify/naca0012_m015_laminar_re5000_o1`).

4.2 Limiter and positivity

The default limiter is Venkatakrishnan's [6],

$$\phi = \min_{f \in \partial c} \frac{1}{\Delta^-} \frac{(\Delta^{+2} + \epsilon^2)\Delta^- + 2\Delta^{-2}\Delta^+}{\Delta^{+2} + 2\Delta^{-2} + \Delta^+\Delta^- + \epsilon^2}, \tag{18}$$

where $\Delta^- = \nabla W_c \cdot (\mathbf{x}_f - \mathbf{x}_c)$ is the unlimited increment and $\Delta^+ = W^{\max} - W_c$ or $W^{\min} - W_c$ according to the sign of Δ^-, with $W^{\max}, W^{\min}$ taken over the cell and its stencil (face neighbours plus boundary states). The smoothing parameter is $\epsilon^2 = (Kh_c)^3\, s_v^2$ with $h_c = \sqrt{V_c}$, $K = 5$ by default and s_v a freestream reference magnitude of variable v; the extra s_v^2 makes ϵ^2 dimensionally consistent with $\Delta^{\pm 2}$, which the usual form implicitly assumes by working with $O(1)$ variables. Barth–Jespersen limiting [5] ($\phi = \min(1, \Delta^+/\Delta^-)$) is available with `--limiter barth`, and `--limiter none` disables limiting for verification purposes.

Positivity is protected in two places. After reconstruction, any face state with $\rho < 10^{-10}\rho_\infty$ or $p < 10^{-10}p_\infty$ is replaced by the first-order cell-centre state and counted. After the implicit update, each cell is checked and its update is halved up to eight times until ρ and p are positive; if no admissible step is found the cell keeps its previous state. Both counters are reported.

4.3 Inviscid flux

Three approximate Riemann solvers are implemented in `src/physics/FluxInviscid.hpp`.

Roe with Harten–Yee entropy fix [1, 2]. The Roe-averaged state is formed with the usual $\sqrt{\rho}$ weighting, the jump is decomposed into the four waves

$$\alpha_1 = \frac{\Delta p - \bar{\rho}\bar{a}\,\Delta u_n}{2\bar{a}^2}, \quad \alpha_2 = \Delta\rho - \frac{\Delta p}{\bar{a}^2}, \quad \alpha_3 = \bar{\rho}\,\Delta u_t, \quad \alpha_4 = \frac{\Delta p + \bar{\rho}\bar{a}\,\Delta u_n}{2\bar{a}^2}, \tag{19}$$

and the flux is $\frac{1}{2}(\mathbf{F}_L + \mathbf{F}_R) - \frac{1}{2}\sum_k |\lambda_k|\alpha_k \mathbf{r}_k$. The entropy fix $|\lambda| \mapsto \frac{1}{2}(\lambda^2/\delta + \delta)$ for $|\lambda| < \delta$, $\delta = \varepsilon(|u_n| + a)$ with $\varepsilon = 0.1$, is applied to the two acoustic (genuinely nonlinear) fields only, not to the linearly degenerate entropy and shear waves.

HLLC [3]. Wave speeds use the Einfeldt–Batten estimate [4] $S_L = \min(u_{n,L} - a_L, \bar{u}_n - \bar{a})$, $S_R = \max(u_{n,R} + a_R, \bar{u}_n + \bar{a})$ with the Roe average for the bar quantities, and the contact speed follows from the normal-momentum balance.

Rusanov / local Lax–Friedrichs. Provided with a configurable dissipation scale (the case file's `rusanov_dissipation_scale`).

HLLC is the production default, for positivity rather than for accuracy: its wave-speed estimate keeps the intermediate states physical without help, where the Roe flux relies on the entropy fix and, at Mach 2, needed the shock fix below before it would converge at all. The Roe implementation is verified to be algebraically exact — a unit test constructs a stationary normal shock from the Rankine–Hugoniot relations and checks that the Roe flux reproduces the common flux $\mathbf{F}_L = \mathbf{F}_R$ to 10^{-9}, which only holds if the Roe average and the wave strengths are correct — and table 6 shows that the two agree on every case tried, from 4×10^{-7} relative in C_D on the cylinder to 7×10^{-3} at Mach 2.

Multidimensional shock fix. Resolving the contact wave is what makes Roe and HLLC accurate for boundary layers, and it is also what makes them vulnerable to the carbuncle instability: inside a captured strong shock, a contact-resolving flux supplies almost no dissipation *along* the front, so a transverse perturbation of the shock is not damped. Left alone this is not a cosmetic problem. On the Mach 2 aerofoil it drove the leading-edge surface pressure to $C_p = 2.120$ against the Rayleigh–Pitot ceiling of 1.657 — a supersonic body cannot produce more stagnation pressure than a normal shock — and broke the symmetry of a symmetric problem badly enough to give $|C_L| = 4.6 \times 10^{-3}$ where the exact value is zero (table 3).

The cure implemented here blends the flux towards Rusanov, but only on the faces that feed the instability. Three dimensionless factors multiply:

$$\omega = \min\Big(1,\ k\ \underbrace{\left[\tfrac{M-0.95}{0.10}\right]_0^1}_{\text{supersonic}}\ \underbrace{\left[\tfrac{\sigma-0.25}{0.50}\right]_0^1}_{\text{shock strength}}\ \underbrace{(1-\alpha^2)}_{\text{transverse}}\Big), \qquad \mathbf{\Phi} = (1-\omega)\mathbf{\Phi}_{\text{HLLC}} + \omega\,\mathbf{\Phi}_{\text{Rusanov}}, \tag{20}$$

where $[\cdot]_0^1$ clips to $[0, 1]$, $M = \max(M_L, M_R)$, $\sigma = |\nabla p|\,|\mathbf{x}_R - \mathbf{x}_L|/p$ is the relative pressure jump across the face, and $\alpha = |\hat{\mathbf{n}} \cdot \nabla p|/|\nabla p|$ measures how the face is oriented relative to the shock. A face that the shock crosses head-on has $\alpha = 1$ and keeps the *full* contact-resolving flux, so the shock itself stays sharp; a face lying in the front has $\alpha = 0$ and is damped. The supersonic factor switches the fix off entirely in subsonic flow, where a shock cannot exist — without it the pressure-gradient sensor also fires on the stagnation region of a stretched mesh, and the $M_\infty = 0.15$ and cylinder results change in the fourth digit for no reason. With it, the cylinder and the $M_\infty = 0.15$ laminar aerofoil rerun with `--shock-fix 0` reproduce their production drag to 2.3×10^{-6} relative — and since those runs use 4 ranks against the production 8, that residual is the rank-count difference of table 13 rather than the fix, which on those cases never activates. The transonic $M_\infty = 0.8$ case, which does contain a shock, moves by 1.6×10^{-3}. The two Mach 2 cases are the ones the fix is for.

The pressure gradient is the same halo-exchanged least-squares gradient used for the reconstruction, and every quantity in eq. (20) is face-averaged, so the blended flux is identical on the two ranks that share a cut face and the fix does not disturb partition independence. The strength k is the one calibration constant; the default $k = 6$ was fixed by requiring that the computed stagnation pressure not exceed the Rayleigh–Pitot value by more than a few per cent, and `--shock-fix 0` disables the fix. Because the sensor is off wherever the flow is subsonic or the pressure gradient is smooth, it leaves the five subsonic cases untouched.

Table 3: Multidimensional shock fix on the Mach 2 aerofoil. C_L is exactly zero for this symmetric section at zero incidence, and $C_p^{\max}$ cannot exceed the Rayleigh–Pitot value, so both columns are absolute error measures. The last column is the rms difference between the upper and lower surface C_p distributions, which is likewise zero for the exact solution. Rusanov does not resolve the contact wave and is therefore immune to the instability, at the cost of dissipation everywhere else; the shock fix recovers most of its robustness while keeping HLLC away from the shock. Source: `results/` and `studies/verify/`.

variant	C_D	$\lvert C_L\rvert$	$C_p^{\max}$	upper/lower C_p rms	steps
HLLC + shock fix (production)	0.09192	8.0×10^{-4}	1.768	3.1×10^{-2}	2554
HLLC, fix disabled	0.08789	4.6×10^{-3}	2.120	2.5×10^{-1}	15287
Rusanov	0.09202	6.0×10^{-5}	1.645	3.7×10^{-3}	1965
Rayleigh–Pitot ceiling			1.657	*exact upper bound*	

4.4 Viscous flux

Face gradients of (u, v, T) use the standard directional-derivative correction, which removes the odd–even decoupling of a plain average and remains consistent on skewed cells:

$$\nabla\varphi_f = \overline{\nabla\varphi} + \left[\frac{\varphi_R - \varphi_L}{|\mathbf{d}|} - \overline{\nabla\varphi}\cdot\hat{\mathbf{e}}\right]\hat{\mathbf{e}}, \qquad \mathbf{d} = \mathbf{x}_R - \mathbf{x}_L,\ \hat{\mathbf{e}} = \mathbf{d}/|\mathbf{d}|, \tag{21}$$

with $\overline{\nabla\varphi} = \frac{1}{2}(\nabla\varphi_L + \nabla\varphi_R)$. The temperature gradient is obtained from the reconstructed ρ and p gradients, $\nabla T = (\nabla p - (p/\rho)\nabla\rho)/(\rho R)$. The stress tensor eq. (5) and heat flux are then evaluated at the face and contracted with the normal. On a boundary face the same formula is used with φ_R replaced by the boundary value and $\mathbf{d} = \mathbf{x}_f - \mathbf{x}_L$, so the wall-normal derivative is $(\varphi_w - \varphi_c)/|\mathbf{x}_f - \mathbf{x}_c|$ — the usual one-sided wall gradient.

5 Boundary Conditions

Each boundary type provides two states (`src/physics/BoundaryCondition.hpp`): a *ghost* state handed to the same approximate Riemann solver used on interior faces (weak imposition), and a *boundary value* that is the physical state on the wall, used for the least-squares stencil, the viscous wall flux, the force integration and the surface output. Keeping these two separate is what allows `surface.csv` to report genuine boundary-condition values.

Farfield. A characteristic condition. If the normal Mach number satisfies $|M_n| \geq 1$ the ghost state is the freestream (inflow) or the interior state (outflow). Otherwise the incoming and outgoing Riemann invariants $R^{\pm} = u_n \pm 2a/(\gamma - 1)$ are combined,

$$u_{n,b} = \tfrac{1}{2}(R^+_{\text{int}} + R^-_{\infty}), \qquad a_b = \tfrac{\gamma-1}{4}(R^+_{\text{int}} - R^-_{\infty}), \tag{22}$$

and entropy and tangential velocity are taken from the upwind side (∞ if $u_{n,b} \leq 0$, interior otherwise), giving $\rho_b = (a_b^2/\gamma s)^{1/(\gamma-1)}$, $p_b = s\rho_b^\gamma$ with $s = p/\rho^\gamma$ of the upwind side. A unit test checks that a freestream interior state reproduces the freestream exactly for arbitrary normals, so the farfield is freestream-preserving.

Slip wall. The ghost state mirrors the normal velocity, $\mathbf{u}_g = \mathbf{u}_i - 2(\mathbf{u}_i \cdot \hat{\mathbf{n}})\hat{\mathbf{n}}$, with ρ and p continuous. With this mirrored pair *every* implemented flux function returns a wall flux of the form

$$\boldsymbol{\Phi} = [0,\ p_w n_x,\ p_w n_y,\ 0]^T, \tag{23}$$

i.e. zero mass flux, zero energy flux and a purely normal momentum flux; a unit test asserts exactly that structure for Roe, HLLC and Rusanov, for both wall types and several normal directions. The effective wall pressure depends on the flux function. For Roe it is

$$p_w = p_i + \rho_i u_{n,i}^2 + \rho_i \bar{a}\, u_{n,i}, \qquad \bar{a}^2 = a_i^2 + \tfrac{\gamma-1}{2} u_{n,i}^2, \tag{24}$$

which a second unit test verifies as an algebraic identity; HLLC gives the same expression when $u_{n,i} \geq 0$ and $p_w = p_i + \rho_i a_i u_{n,i}$ when $u_{n,i} < 0$, because its wave-speed estimate then selects a different left state. In both cases the correction is proportional to $u_{n,i}$ and therefore acts to drive the normal velocity to zero, and both reduce to $p_w = p_i$ at convergence. The reported boundary value keeps the tangential velocity, $\mathbf{u}_b = \mathbf{u}_i - (\mathbf{u}_i \cdot \hat{\mathbf{n}})\hat{\mathbf{n}}$, so slip-wall rows in `surface.csv` have zero normal velocity and non-zero tangential velocity.

No-slip adiabatic wall. The *convective* part of a solid-wall flux carries no mass, so the ghost state again mirrors only the normal velocity and eq. (23) applies unchanged. Reversing the tangential velocity as well — a tempting way to "impose" no slip — would leave a spurious convective tangential momentum flux $\rho u_t u_n$ at the wall; this was measured during development and is why the mirrored-normal form is used. The no-slip condition is imposed through the viscous stress instead, which is built directly from the wall-normal velocity derivative as described below. The reported boundary value is $\mathbf{u}_b = \mathbf{0}$ with $p_b = p_i$ and $T_b = T_i$ (adiabatic); it enters the least-squares stencil, the force integration and the surface output. The adiabatic condition is enforced by dropping the heat-flux term entirely, so $\mathbf{q} \cdot \hat{\mathbf{n}} = 0$ identically, and the viscous energy flux vanishes because $\mathbf{u}_b = \mathbf{0}$: the wall exchanges only momentum with the fluid. Wall rows in `surface.csv` therefore report exactly zero velocity and zero Mach number; the adjacent cell-centre values are written separately to `surface_cellcenter.csv` so that the two can never be confused.

Wall stress. At a steady no-slip wall the viscous traction is *purely tangential*, and the solver builds it that way rather than assembling it from the general face-gradient formula. The reason is exact: with $\mathbf{u} = \mathbf{0}$ everywhere on the wall, the tangential derivative of the tangential velocity vanishes, and continuity $\rho\, \nabla\cdot\mathbf{u} + \mathbf{u} \cdot \nabla\rho = 0$ with $\mathbf{u} = \mathbf{0}$ gives $\nabla\cdot\mathbf{u} = 0$ there. In wall coordinates that leaves $\tau_{nn} = \tau_{tt} = 0$ and only

$$\boldsymbol{\tau} \cdot \hat{\mathbf{n}} = \mu \frac{\partial u_t}{\partial n} \hat{\mathbf{t}} = -\frac{\mu}{d_n}\, \mathbf{u}_{t,c}, \qquad d_n = (\mathbf{x}_f - \mathbf{x}_c) \cdot \hat{\mathbf{n}}, \tag{25}$$

with $\mathbf{u}_{t,c}$ the tangential cell-centre velocity. Assembling the same quantity from the general formula instead lets the tangential part of the cell-centre least-squares gradient leak into the discrete $\nabla\cdot\mathbf{u}$, and on cells whose aspect ratio reaches 2821 that leakage is not small: during development it produced a spurious normal traction worth about two thirds of the drag of the Mach 2 laminar case, where the exact value is zero. With eq. (25) the normal viscous traction is identically zero, and `sanity_checks.json` asserts that on every viscous case. Farfield faces still use the general formula.

Force decomposition. With $\hat{\mathbf{n}}$ the outward normal of the *fluid* domain, the traction the fluid exerts on the body is $\mathbf{T} = p\hat{\mathbf{n}} - \boldsymbol{\tau}\cdot\hat{\mathbf{n}}$. The viscous traction is split into its normal and tangential parts, and only the tangential part — the true skin friction — is reported as `viscous_drag`/`viscous_lift`:

$$\boldsymbol{\tau}_t = \boldsymbol{\tau}\cdot\hat{\mathbf{n}} - \big[(\boldsymbol{\tau}\cdot\hat{\mathbf{n}})\cdot\hat{\mathbf{n}}\big]\hat{\mathbf{n}}, \qquad C_f = \frac{-\boldsymbol{\tau}_t\cdot\hat{\mathbf{t}}}{q_\infty}, \tag{26}$$

where $\hat{\mathbf{t}}$ is the unit surface tangent oriented so that $\hat{\mathbf{t}}\cdot\hat{\mathbf{d}} \geq 0$. With this convention $C_f > 0$ means wall shear acting downstream (attached flow) and $C_f < 0$ marks reversed flow, on both sides of a symmetric body. The small normal viscous traction is reported separately (`final_normal_viscous_drag` in `metadata.json`) and is included in C_D so that the total force remains exact; it is below 0.2% of C_D in every viscous case here.

6 Implicit and Transient Time Integration

6.1 Linearisation and the LU-SGS relaxation

Both the steady and the transient path solve, at every nonlinear step,

$$\Big[\frac{V_c}{\Delta\tau_c}\mathbf{I} + a_0\frac{V_c}{\Delta t}\mathbf{I} + \frac{\partial\mathbf{R}}{\partial\mathbf{U}}\Big]\Delta\mathbf{U} = -\mathbf{R}^*, \tag{27}$$

with $a_0 = 0$ for steady marching. The Jacobian is approximated face by face with flux-vector splitting,

$$\frac{\partial\boldsymbol{\Phi}_f}{\partial\mathbf{U}_c} = \tfrac{1}{2}\big(A_c\cdot\hat{\mathbf{n}} + \lambda_f\mathbf{I}\big)\ell_f, \qquad \frac{\partial\boldsymbol{\Phi}_f}{\partial\mathbf{U}_j} = \tfrac{1}{2}\big(A_j\cdot\hat{\mathbf{n}} - \lambda_f\mathbf{I}\big)\ell_f, \tag{28}$$

where λ_f is the face spectral radius

$$\lambda_f = \overline{|u_n| + a} + 2\kappa_f, \qquad \kappa_f = \max\Big(\tfrac{4}{3}, \tfrac{\gamma}{\mathrm{Pr}}\Big)\frac{\mu_f}{\rho_f|\mathbf{d}_{LR}|}. \tag{29}$$

Because $\sum_f \hat{\mathbf{n}}_f\ell_f = 0$ for a closed cell, the A_c contribution cancels and the diagonal collapses to the scalar

$$D_c = \frac{V_c}{\Delta\tau_c} + a_0\frac{V_c}{\Delta t} + \tfrac{1}{2}\sum_{f\in\partial c}\lambda_f\ell_f, \tag{30}$$

so no 4×4 block ever has to be stored or inverted. The off-diagonal product is evaluated matrix-free as a true nonlinear flux difference,

$$\tfrac{1}{2}\Big[\mathbf{F}(\mathbf{U}_j + \Delta\mathbf{U}_j)\cdot\hat{\mathbf{n}} - \mathbf{F}(\mathbf{U}_j)\cdot\hat{\mathbf{n}} - \lambda_f\Delta\mathbf{U}_j\Big]\ell_f, \tag{31}$$

with a guard that falls back to the dissipative term alone if $\mathbf{U}_j + \Delta\mathbf{U}_j$ is not a physical state. The system is then relaxed with symmetric Gauss–Seidel sweeps (forward followed by backward) over the rank-local cells, which are renumbered with reverse Cuthill–McKee inside each partition to shorten the dependency chains. This is the classical Yoon–Jameson LU-SGS [7] in matrix-free form; the spectral-radius estimates and the local time step follow the standard forms collected by Blazek [9].

Local pseudo-time steps come from the same spectral radii,

$$\Delta\tau_c = \mathrm{CFL}\,\frac{V_c}{\sum_f(|u_n|+a)_f\ell_f + 4\sum_f \kappa_f\ell_f}, \tag{32}$$

and the CFL follows the case schedule geometrically, $\mathrm{CFL}(n) = \mathrm{CFL}_0(\mathrm{CFL}_{\max}/\mathrm{CFL}_0)^{\min(1,n/n_{\mathrm{ramp}})}$.

6.2 Steady path

One nonlinear step per outer iteration; the inner iterations are the LU-SGS sweeps, controlled by the case values `min_inner_iterations`, `max_inner_iterations` and `inner_residual_reduction_target` applied to the *linear* residual $\|D\Delta\mathbf{U} + \text{off-diag} + \mathbf{R}\|/\|\mathbf{R}\|$, which is evaluated explicitly after each sweep. The residual is measured at the state at the beginning of each outer step, and the loop is arranged so that the final row of `residuals.csv` and `forces.csv` corresponds exactly to the state written to `surface.csv`, `field_final.vtu` and `restart_final.bin`.

6.3 Transient path: dual-time BDF2

The Reynolds 200 cylinder uses a genuine two-level dual-time loop [8]. The outer loop advances physical time with

$$\mathbf{R}^*(\mathbf{U}^{n+1}) = V\frac{3\mathbf{U}^{n+1} - 4\mathbf{U}^n + \mathbf{U}^{n-1}}{2\Delta t} + \mathbf{R}(\mathbf{U}^{n+1}) = \mathbf{0}, \tag{33}$$

and the inner loop drives $\|\mathbf{R}^*\|$ down by solving eq. (27) with $a_0 = 3/2$ and updating $\mathbf{U}^{n+1,k+1} = \mathbf{U}^{n+1,k} + \Delta\mathbf{U}$. $\mathbf{U}^n$ and $\mathbf{U}^{n-1}$ are *frozen* for the entire inner loop; the histories are shifted only after the physical step is accepted (`RunLoop.cpp`, `runTransient`). The very first physical step uses backward Euler because no $\mathbf{U}^{n-1}$ exists yet, which is the standard self-starting variant and does not affect the global second-order accuracy. A trapezoidal variant is implemented and selectable through the case file.

Inner convergence is measured on the *total* transient residual eq. (33), exactly as the case file requests (`inner_residual_norm: total_spatial_plus_physical_time`): the loop stops when $\|\mathbf{R}^*\|_2/\|\mathbf{R}^{*,0}\|_2 \le \varepsilon_{\mathrm{in}}$ and at least `min_inner_iterations` updates have been performed, and gives up at `max_inner_iterations`. Forces and residuals are sampled once per *physical* step. The realised statistics — minimum, mean and maximum inner iterations, number of target misses, converged fraction and the last residual ratio — are written to `metadata.json` and collected in table 10.

6.4 Parameters actually used

Table 9 lists the production controls of every submitted run alongside any extra command-line options. Every case uses the CFL schedule, step budget, inner-iteration bounds and inner target of its own JSON file. Two mechanisms go beyond the literal case file, both documented and neither of them loosening the computation:

1. **Limiter freezing (all steady cases, automatic).** The min / max stencil of any Barth-type limiter is a non-differentiable function of the solution, and on the shock-containing cases this produces a residual limit cycle: the residual stalls at a few times 10^{-3} of its initial value

while the limiter pattern flips between two states (fig. 16). The solver therefore holds the limiter values fixed from pseudo-time step $3\,n_{\mathrm{ramp}}$ onwards, i.e. two full CFL-ramp lengths after the ramp has finished. This is a *uniform* rule — it is derived from the case file's own `pseudo_cfl_ramp_steps` and needs no per-case flag — and it is a freeze, not a switch-off: the frozen values keep multiplying the reconstruction in eq. (17), so the scheme remains limited everywhere it was limited at the freeze step. On the smooth subsonic cases the freeze step is never reached, because they converge first. The behaviour can be disabled with `--freeze-limiter-step 0` and is recorded in `metadata.json`.

2. **Reynolds 200 cylinder:** `--cfl-scale 30 --inner-target 1e-4`. The supplied pseudo-time CFL of 1.0 makes the dual-time diagonal $V/\Delta\tau$ dominate the physical-time term $1.5V/\Delta t$ by a factor $\Delta t/(1.5\Delta\tau) \sim 30$ in the near-wall cells, so the inner loop degenerates into a slow pseudo-time march and needs 147 iterations per physical step on average. Raising the pseudo-time CFL changes only the convergence *rate* of the inner problem, never its converged solution. Table 4 separates the two effects. At the *supplied* inner target of 10^{-3}, raising the pseudo-CFL from 1 to 100 changes C_D by 0.31% and the cost by an order of magnitude: a relative reduction of the first inner residual does not by itself pin the answer, so the supplied target leaves a visible dual-time error. Tightening the target to 10^{-4} removes almost all of that spread — pseudo-CFL 30 and 100 then agree to 1.6×10^{-4} — and 10^{-5} removes the rest. Measured against that most tightly converged row, the literal supplied setting is 0.58% off and the production setting 0.031% off, while costing 19.6 s against 3.5 s and 147 against 30 inner iterations per physical step. As an independent check on a much longer interval, running the case to $t = 20$ with the literal supplied controls gives $C_D = 0.937914$ against 0.937905 with the production controls — a relative difference of 9.78×10^{-6} — while taking 605 s instead of 176 s and 122 instead of 33 inner iterations per physical step (`studies/verify/cylinder_m010_laminar_re200_*`). The physical time step ($\Delta t = 0.01$), the final time ($t_f = 300$), the inner-iteration bounds (5–1000) and the frozen-history rule are all exactly as supplied.

Table 4: Effect of the pseudo-time CFL and of the inner residual target on the dual-time solution of the Reynolds 200 cylinder. All rows use the same BDF2 discretisation and the same physical time step $\Delta t = 0.01$, are run on 4 MPI ranks, and are compared at the same physical time. The last column is the difference from the most tightly converged row, i.e. an estimate of the remaining dual-time error. The supplied setting (pseudo-CFL 1, target 10^{-3}) is further from the converged dual-time answer than the production setting (pseudo-CFL 30, target 10^{-4}) and costs several times more. Produced by `scripts/run_cfl_study.sh`; source `studies/cflstudy/`.

pseudo-CFL	inner target	mean inner its	wall time [s]	$C_D(t = 0.5)$	$\lvert\Delta C_D\rvert$ vs. tightest
1	1×10^{-3}	147.1	19.6	1.063106	6.18×10^{-3}
10	1×10^{-3}	22.2	2.7	1.062058	5.13×10^{-3}
30	1×10^{-3}	14.7	1.8	1.061034	4.11×10^{-3}
100	1×10^{-3}	12.8	1.7	1.059825	2.90×10^{-3}
30	1×10^{-4}	29.5	3.5	1.057253	3.25×10^{-4}
100	1×10^{-4}	25.8	3.3	1.057089	1.61×10^{-4}
30	1×10^{-5}	63.3	9.6	1.056928	0

7 MPI Parallelisation and METIS Partitioning

7.1 Preprocessing and distribution

Rank 0 reads the CGNS file, builds the face topology and the cell adjacency (dual) graph in CSR form, and calls `METIS_PartGraphKway` [11] with `METIS_OBJTYPE_CUT`, contiguity requested and a fixed seed, so the partition is deterministic and reproducible. The cells of each part are then renumbered with reverse Cuthill–McKee. For every rank r the preprocessing step assembles the owned cells, the one-layer ghost set (face neighbours owned by other ranks, sorted by owner and global id), the local node set and the boundary edges of owned cells, packs them into two flat buffers and sends them with `MPI_Send`. **The global mesh is then released on rank 0 before the solver allocates any state**, and each rank rebuilds its own face topology, geometry and least-squares stencils from its sub-mesh using exactly the same code as the serial path. Consequently no rank ever holds the global mesh or a global conservative state during iterations, which `metadata.json` asserts through `full_mesh_replication_during_iterations` and `full_state_replication_during_iterations`.

Halo descriptors are built without any global data structure: each rank groups its ghosts by owner to form the receive lists, exchanges the requested global ids with those neighbours, and translates them into local owned indices to form the send lists. The only collective involved is an `MPI_Alltoall` on a vector of *integer counts*, once, at setup.

7.2 Communication during iterations

Each residual evaluation performs three neighbour-scoped exchanges with `MPI_Isend`/`MPI_Irecv`/`MPI_Waitall`: the conservative state (4 doubles per ghost cell), then the primitive gradients (8) and the limiter values (4). Exchanging gradients and limiters rather than adding a second ghost layer is deliberate: the gradient and limiter of a ghost cell are computed by the rank that owns its *complete* stencil, so the reconstructed face states — and therefore the residual — are identical to the serial ones regardless of how the mesh is cut. The LU-SGS sweeps exchange $\Delta\mathbf{U}$ after each half sweep, which makes the scheme a hybrid LU-SGS/block-Jacobi: exact within a rank, lagged across rank boundaries. This affects only the iteration path, not the converged answer, which is why the steady force coefficients in table 13 agree across rank counts to the level of the convergence tolerance. Residual and force norms use `MPI_Allreduce`; no `MPI_Allgather` of state or mesh data occurs anywhere in the code.

7.3 Partition quality

Table 14 lists the per-rank owned/ghost counts, neighbour counts and halo sizes for the reference 8-rank NACA run (METIS edge cut 439, load-balance ratio $\max/\text{mean} = 1.0154$). The same information is written for every case to `partition_diagnostics.csv/.json`, and the actual partition is visualised in the `*_partition.png` figures, which colour each cell by the rank that owned it during the run.

7.4 Rank-independent output

Every file a run produces is written by rank 0 from globally reduced or globally ordered data, so its contents do not depend on how the mesh happened to be split:

- `residuals.csv` and `forces.csv` carry values that have already been through `MPI_Allreduce`, so they are rank-independent by construction.

- `surface.csv` and `surface_cellcenter.csv` are gathered and then sorted by (patch, azimuth about the body centre), a key that has nothing to do with the partition. `tools/mpi_study.py` compares them across the rank study and writes the result to `report/rank_independence.json`: in all 12 comparisons the row count, the row order and every geometric column are *identical* (0 failures), so row k of the file is the same boundary face at every rank count. The solution columns cannot be bit-identical, because different rank counts take slightly different LU-SGS paths and stop at different steps; they agree to 7.2×10^{-4} at the 99th percentile. The single worst disagreement, 9.4×10^{-3}, is in `cp` at $x = 1.0052$ — the sharp trailing edge of the aerofoil, where the inviscid solution is genuinely singular (section 2.3) and where the residual also concentrates. It does not reach the integrated forces, which agree to 1.1×10^{-5} absolute.

- `field_final.vtu` is assembled deterministically: the cells are sorted by global cell id and the unique mesh points are numbered by global node id, so the point block, the connectivity block and the cell ordering are identical at every rank count and two runs can be compared row for row. The one deliberate exception is the `RankId` field, whose whole purpose is to record the partition. `tests/mpi_consistency.sh` asserts all of this as a regression test — identical point and connectivity blocks against the single-rank run, cells in ascending global id, every point referenced by a cell and every field finite — because a defect in the rank-0 gather is invisible in the CSV outputs and would otherwise only be noticed by looking at a picture.

- `restart_final.bin` is written in global cell order, which is what lets a state written on 8 ranks be reloaded on 4 (section 9).

8 Verification

Verification is independent of the benchmark cases and is produced by `cfd2d verify`, whose output is `report/verification.json`.

8.1 Unit tests

`ctest` runs two tests. The first is a `doctest` suite of 20 test cases (270 assertions, 0 failing; `report/unit_tests.log`) covering polygon area/centroid and edge normals, the equation-of-state round trip, consistency of all three Riemann solvers ($\mathbf{\Phi}(\mathbf{W}, \mathbf{W}) = \mathbf{F}(\mathbf{W})$), the exact upwind limit for supersonic states, exactness of the Roe flux across a stationary Rankine–Hugoniot shock, the algebraic wall-flux identity eq. (23) for both wall types, freestream preservation of the farfield condition, and the stress tensor and adiabatic viscous flux. The second, `tests/mpi_consistency.sh`, is a parallel regression test: it runs the same case on 1, 2 and 4 ranks and checks partition completeness, non-empty and symmetric halos, agreement of the global residual and forces, and the determinism of the field file described in section 7.4.

8.2 Freestream preservation and linear exactness on the benchmark mesh

On the NACA0012 mesh, a uniform freestream produces a residual of 2.73×10^{-10} (normalised as in eq. (11)) in all cells whose stencil does not touch a boundary — machine zero. (A uniform state is *not* a solution of the problem with a body in it, so the wall cells and their immediate neighbours legitimately carry a residual; those two layers are excluded.) Feeding an exactly linear primitive field through the same code path recovers the gradient to a relative error of 7.97×10^{-8} and the reconstructed face values to 6.66×10^{-16}, confirming that the least-squares reconstruction is exact for linear data, which is the algebraic prerequisite for second-order accuracy.

8.3 Order of accuracy

The formal order is measured with the method of manufactured solutions [10] on a sequence of internally generated meshes of $[0,1]^2$. Each mesh mixes quadrilaterals and triangles in a chequerboard pattern and is a smooth analytic distortion of a Cartesian grid, so it is genuinely unstructured and non-uniform. The manufactured state is a C^∞, compactly supported perturbation of a uniform $M \approx 0.42$ flow; because the support does not reach the boundary, the manufactured state coincides with the freestream there and the ordinary characteristic farfield condition is exact, so the measured error is purely the interior discretisation error. The flux divergence of the manufactured state is added as a source term, the steady problem is converged by eleven orders, and the solution is compared with *cell averages*, not point values, obtained by the same three-point edge-midpoint quadrature on each sub-triangle that produces the source term. Two consequences are worth stating plainly. The quadrature is exact only for quadratics, so it contributes its own error — measured at 1.6×10^{-4} on the coarsest level and 1.3×10^{-7} on the finest, i.e. converging faster than the discretisation error it is used to measure, but not zero. And the flux divergence that forms the source term is evaluated by second-order central differences with a step of 10^{-5} rather than symbolically; its worst-case error over the levels used is 2×10^{-8}, three orders below the finest discretisation error in table 5. What matters is that the *same* quadrature is used for the source term and for the error: a source term built from centroid point values would be inconsistent with a cell-averaged error at $\mathcal{O}(h^2)$ and would corrupt the measured order.

Table 5: Manufactured-solution convergence study. The error is the L_1/L_2 norm of the difference between the converged discrete solution and the exact cell averages of the manufactured primitive state; "local order" is $\log(e_{i-1}/e_i)/\log(h_{i-1}/h_i)$ between successive levels. Source: `report/verification.json`.

scheme	cells	mean h	L_1 error	L_2 error	local order
2nd order, unlimited	384	0.0542	2.950×10^{-3}	3.776×10^{-3}	
	1536	0.0271	5.276×10^{-4}	7.326×10^{-4}	2.48
	6144	0.0136	1.110×10^{-4}	1.658×10^{-4}	2.25
	24576	0.0068	2.452×10^{-5}	3.650×10^{-5}	2.18
2nd order, Venkatakrishnan	384	0.0542	2.967×10^{-3}	3.809×10^{-3}	
	1536	0.0271	5.397×10^{-4}	7.470×10^{-4}	2.46
	6144	0.0136	1.268×10^{-4}	1.863×10^{-4}	2.09
	24576	0.0068	4.423×10^{-5}	6.782×10^{-5}	1.52
1st order	384	0.0542	1.181×10^{-2}	1.764×10^{-2}	
	1536	0.0271	6.339×10^{-3}	9.607×10^{-3}	0.90
	6144	0.0136	3.389×10^{-3}	5.196×10^{-3}	0.90
	24576	0.0068	1.755×10^{-3}	2.711×10^{-3}	0.95

Table 5 and fig. 1 give the result. The unlimited second-order scheme converges at order 2.30, the same scheme with the Venkatakrishnan limiter active at 2.03, and the first-order scheme at 0.92. The second-order figure exceeds two: on a smoothly mapped mesh with a smooth solution the leading truncation terms partially cancel, and finite-volume schemes are known to superconverge in that situation, so the measurement should be read as "at least second order" rather than as a sharp exponent. The manufactured field is smooth, so the limiter is close to inactive and the two second-order variants agree; this is exactly what a well-behaved limiter should do, and it is direct evidence that the limiter is not degrading the scheme in smooth regions on meshes of this resolution. On a mesh fine enough for $\epsilon^2 = (Kh)^3$ to fall below $\Delta^{\pm 2} \sim h^2$ the limiter would become Barth-like and the order would drop; that is a property of the Venkatakrishnan family rather than

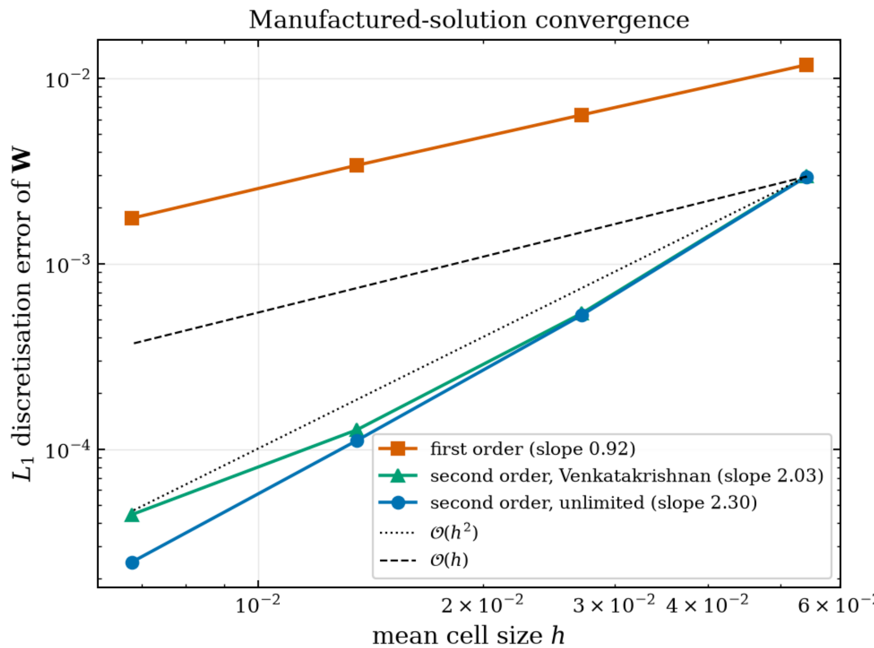


Figure 1: Manufactured-solution discretisation error against mean cell size for the first-order, limited second-order and unlimited second-order schemes, with $\mathcal{O}(h)$ and $\mathcal{O}(h^2)$ reference slopes.

of the reconstruction, and it is why the constant K matters (section 14).

For completeness the truncation error of the residual operator is also reported in `verification.json`. On non-uniform unstructured meshes it converges more slowly than the solution error — the classical supraconvergence behaviour of finite-volume schemes [17] — so the solution-error study above, not the truncation error, is the meaningful measure of the formal order.

8.4 Flux-scheme cross-check

Table 6 compares the production HLLC results with the independently implemented Roe flux with Harten–Yee entropy fix on three cases, one subsonic, one supersonic and one viscous. The two solvers share only the mesh, the reconstruction and the time integration; the flux itself is a separate derivation. They agree to 4×10^{-7} relative in C_D on the cylinder, 5×10^{-3} on the subsonic aerofoil — where C_D is the 10^{-3} spurious drag of a case whose exact drag is zero, so the absolute agreement is 5×10^{-6} — and 7×10^{-3} at Mach 2, where the answer is set by a captured bow shock. A third, structurally different scheme, Rusanov, lands within 0.1% of both at Mach 2 (table 3). Three independent flux functions agreeing is the strongest evidence in this report that the residual assembly around them is correct.

Both contact-resolving schemes need the shock fix of eq. (20) at Mach 2. Without it the Roe run does not converge at all; with it, it converges in 2404 steps with 0 positivity fallbacks, and the same fix takes HLLC from a 28% stagnation-pressure overshoot to 7%. That one mechanism explains both schemes' behaviour is itself a check on the diagnosis.

Table 6: Independent cross-check of the two contact-resolving approximate Riemann solvers: the production HLLC runs against runs of the same cases with the Roe flux and its Harten–Yee entropy fix, on identical meshes and with identical convergence criteria and shock fix. Source: `results/` and `studies/verify/`.

case	flux	C_L	C_D	residual orders	steps	status
NACA0012 $M_\infty = 0.15$ inviscid	hllc	+0.000147	0.001025	4.32	1864	converged
	roe + Harten–Yee	+0.000053	0.001021	4.29	1894	converged
	relative C_D difference		4.53×10^{-3}			
NACA0012 $M_\infty = 2.0$ inviscid	hllc	+0.000797	0.091919	3.07	2554	converged
	roe + Harten–Yee	+0.000500	0.091265	3.00	2404	converged
	relative C_D difference		7.11×10^{-3}			
cylinder $M_\infty = 0.1$ laminar $Re = 20$	hllc	+0.000643	2.017617	5.37	2806	converged
	roe + Harten–Yee	+0.000654	2.017618	5.37	2805	converged
	relative C_D difference		4.12×10^{-7}			

9 Output, Reproducibility and Sanity Checks

Every case directory follows the benchmark's `cfd_solver_agentic_benchmark/OUTPUT_CONTRACT.md`: `metadata.json`, `partition_diagnostics.csv/.json`, `residuals.csv`, `forces.csv`, `surface.csv`, `field_final.vtu`, `restart_final.bin`, `stdout.log` and `run_status.json`, plus three diagnostic extras: `inner_history.csv` (per-step inner-iteration count, residual ratio and convergence flag), `surface_cellcenter.csv` (adjacent cell-centre values next to the wall values) and, for the transient case, `fields/` with the requested snapshots every $\Delta t_{\text{out}} = 1$.

The field file is a standard XML VTK unstructured grid with a raw appended data section containing density, both velocity components, velocity magnitude, pressure, Mach number, temperature, total energy, vorticity, the owning MPI rank and the scaled residual magnitude. It is assembled on rank 0 *after* the iteration loop by gathering the rank-local pieces and de-duplicating nodes by global id; this is an output-time operation and does not violate the no-replication requirement during iterations. The restart file stores the conservative state (and, for transient runs, both BDF2 history levels) in global cell order, so a run can be restarted on a different number of ranks. `scripts/run_verify_extra.sh` exercises exactly that: the converged $M_\infty = 0.15$ inviscid case is written on 8 ranks and reloaded on 4, and the force coefficients recomputed from the reloaded state differ from the originals by exactly zero (`studies/restart/restart_check.json`).

`report/sanity_checks.json` is produced by `tools/sanity_checks.py` and contains 118 checks over 8 cases: positivity of density and pressure and finiteness of every field variable, non-trivial surface C_p variation, zero lift by symmetry for the NACA cases, positive mean drag for the cylinders, unsteady lift for the transient case, the inner-target converged fraction, near-zero wall velocity and non-zero skin friction on no-slip walls, near-zero normal velocity and negligible viscous force columns on slip walls, farfield uniformity, agreement between the last force row and the final state, that the maximum wall pressure coefficient respects the exact inviscid stagnation bound (isentropic below Mach 1, Rayleigh–Pitot above it) within a viscous allowance that scales like $Re^{-1/2}$, that the normal viscous traction on a no-slip wall is identically zero and that the pressure/skin-friction split sums to the reported C_D, and presence of separate Mach and pressure figures — listed in the manifest *and* present on disk — for every case. 0 checks fail. The gate is not advisory: a case whose `run_status.json` claims `converged` or `statistically_periodic` while any of its checks fails is reported as inconsistent and `scripts/check_submission.sh` exits non-zero, so a failure cannot be quietly carried into the submission. Every failing check, if there were one, is named with its measured value in `report/sanity_checks.json`.

`report/figure_manifest.csv` maps every figure to its case, type, plotted variable and source data file; `report/run_manifest.md` lists the exact command, rank count, step count, wall time and status of every submitted run, and the CSV form adds the solver version, the git revision and the numerical settings (flux, reconstruction, limiter, shock-fix strength) each run was produced with, so the provenance of every number is recoverable from the submission alone.

Two conventions in `metadata.json` are worth stating explicitly. `git_revision` is the repository revision captured when CMake was configured, not necessarily the revision of the final commit; `scripts/check_submission.sh` asserts that all eight cases carry the *same* revision, flux, limiter and shock-fix setting, so a submission that mixed results from two builds of the solver would be rejected rather than quietly shipped. The effective pseudo-time CFL of a run is recorded separately from the case-file value (`cfl_scale`, `effective_cfl_initial`, `effective_cfl_max`), so a run that deviates from the case schedule is visible in the metadata and not only in the recorded command line; likewise `limiter_freeze_step` records the step actually used, alongside the command-line option that produced it.

9.1 Build and run commands

The solver is built and run with the following commands; `CFD_EXTERNALS_ROOT` points at the compiled CGNS/HDF5/METIS installation and `CFD_HEADER_ROOT` at the header-only packages.

```
export CFD_EXTERNALS_ROOT=<workspace>/external/cfd_externals/install
export CFD_HEADER_ROOT=<workspace>/external

cmake -S . -B build -DCMAKE_BUILD_TYPE=Release \
      -DCFD_EXTERNALS_ROOT=$CFD_EXTERNALS_ROOT \
      -DCFD_HEADER_ROOT=$CFD_HEADER_ROOT
cmake --build build -j
ctest --test-dir build                    # unit tests + MPI consistency test

mpirun -np 8 --bind-to none --oversubscribe ./build/cfd2d solve \
    --case ../cfd_solver_agentic_benchmark/inputs/cases/naca0012_m015_inviscid.json \
    --output results/naca0012_m015_inviscid --progress-every 2000

scripts/run_all.sh steady        # the seven steady cases at np=8
scripts/run_all.sh transient     # cylinder Re 200 dual-time run
scripts/run_all.sh mpi           # rank-count study, np = 1, 2, 4, 8
scripts/run_all.sh verify        # flux / limiter / dual-time cross-checks
scripts/make_report.sh           # verification, figures, manifests, LaTeX
```

`--bind-to none --oversubscribe` is needed because the benchmark container advertises 64 logical cores but is limited to a 4-CPU cgroup quota; Open MPI's default core pinning collapses performance under such a quota.

Python post-processing runs inside a local virtual environment (`numpy`, `matplotlib`, `scipy`); no Python code participates in the solve.

10 Results

10.1 Summary

Table 7 states, for every case, whether it converged, reached a statistically periodic state, or failed. Table 8 gives the force coefficients with the same verdict repeated alongside each number, table 9 the controls actually used, and table 10 the realised inner-iteration statistics.

Table 7: Run status of all required cases. "Residual orders" is $\log_{10}(\|r\|_2^0/\|r\|_2^{\text{final}})$ with the norm of eq. (11); for the transient case it refers to the total dual-time residual of the last physical step relative to the first.

case	ranks	steps	t_{final}	residual orders	wall time [s]	status
NACA0012 $M_\infty = 0.15$ inviscid	8	1864	0.0	4.32	25	CONVERGED
NACA0012 $M_\infty = 0.80$ inviscid	8	9005	0.0	4.02	146	CONVERGED
NACA0012 $M_\infty = 2.0$ inviscid	8	2554	0.0	3.07	18	CONVERGED
NACA0012 $M_\infty = 0.15$ laminar $Re = 5000$	8	4663	0.0	6.08	43	CONVERGED
NACA0012 $M_\infty = 0.80$ laminar $Re = 5000$	8	8284	0.0	4.76	83	CONVERGED
NACA0012 $M_\infty = 2.0$ laminar $Re = 5000$	8	18135	0.0	5.44	187	CONVERGED
cylinder $M_\infty = 0.1$ laminar $Re = 20$	8	2806	0.0	5.37	44	CONVERGED
cylinder $M_\infty = 0.1$ laminar $Re = 200$	8	30000	300.0	6.29	2311	STAT. PERIODIC

Table 8: Force coefficients. $C_{D,p}$ is the pressure (form) drag and $C_{D,f}$ the tangential skin-friction drag. C_D is the total, which also contains the normal viscous traction; that term is identically zero on no-slip walls by eq. (25) and negligible elsewhere, so $C_D = C_{D,p} + C_{D,f}$ to the digits shown. † time-averaged over the second half of the record.

case	C_L	C_D	$C_{D,p}$	$C_{D,f}$	$C_{m,z}$	status
NACA0012 $M_\infty = 0.15$ inviscid	+0.00015	0.00103	0.00103	0.00000	+0.00006	CONVERGED
NACA0012 $M_\infty = 0.80$ inviscid	-0.00113	0.00854	0.00854	0.00000	-0.00014	CONVERGED
NACA0012 $M_\infty = 2.0$ inviscid	+0.00080	0.09192	0.09192	0.00000	+0.00019	CONVERGED
NACA0012 $M_\infty = 0.15$ laminar $Re = 5000$	-0.00119	0.05339	0.01907	0.03432	-0.00017	CONVERGED
NACA0012 $M_\infty = 0.80$ laminar $Re = 5000$	+0.00163	0.07691	0.04616	0.03075	+0.00016	CONVERGED
NACA0012 $M_\infty = 2.0$ laminar $Re = 5000$	+0.00013	0.13840	0.09706	0.04134	+0.00008	CONVERGED
cylinder $M_\infty = 0.1$ laminar $Re = 20$	+0.00064	2.01762	1.21981	0.79780	-0.00001	CONVERGED
cylinder $M_\infty = 0.1$ laminar $Re = 200$†	+0.00173	1.26606	1.02432	0.24174	+0.00002	STAT. PERIODIC

Table 9: Production numerical controls actually used, read from `metadata.json`. "CFL" and "ramp" are the case pseudo-time schedule, "inner" the inner-iteration bounds and "target" the inner residual target.

case	steps used	CFL	ramp	inner	target	extra options
NACA0012 $M_\infty = 0.15$ inviscid	1864	1–100	2000	3–50	0.01	–
NACA0012 $M_\infty = 0.80$ inviscid	9005	1–100	3000	3–50	0.01	–
NACA0012 $M_\infty = 2.0$ inviscid	2554	0.2–50	5000	3–80	0.01	–
NACA0012 $M_\infty = 0.15$ laminar $Re = 5000$	4663	1–100	5000	3–80	0.01	–
NACA0012 $M_\infty = 0.80$ laminar $Re = 5000$	8284	0.5–80	5000	3–80	0.01	–
NACA0012 $M_\infty = 2.0$ laminar $Re = 5000$	18135	0.2–50	7000	3–100	0.01	–
cylinder $M_\infty = 0.1$ laminar $Re = 20$	2806	1–100	3000	3–80	0.01	–
cylinder $M_\infty = 0.1$ laminar $Re = 200$	30000	1–1	0	5–1000	0.0001	–cfl-scale 30, –inner-target 1e-4

Table 10: Realised inner-iteration statistics. For steady cases the inner iterations are LU-SGS sweeps measured against the linear residual; for the transient case they are dual-time nonlinear iterations measured against the total transient residual.

case	min	mean	max	target	misses	converged fraction
NACA0012 $M_\infty = 0.15$ inviscid	3	5.28	23	0.01	0	100.00%
NACA0012 $M_\infty = 0.80$ inviscid	3	9.10	20	0.01	0	100.00%
NACA0012 $M_\infty = 2.0$ inviscid	3	3.00	3	0.01	0	100.00%
NACA0012 $M_\infty = 0.15$ laminar $Re = 5000$	3	3.82	6	0.01	0	100.00%
NACA0012 $M_\infty = 0.80$ laminar $Re = 5000$	3	4.37	6	0.01	0	100.00%
NACA0012 $M_\infty = 2.0$ laminar $Re = 5000$	3	4.40	6	0.01	0	100.00%
cylinder $M_\infty = 0.1$ laminar $Re = 20$	3	4.92	16	0.01	0	100.00%
cylinder $M_\infty = 0.1$ laminar $Re = 200$	16	22.58	72	0.0001	0	100.00%

10.2 NACA0012, inviscid

All three inviscid aerofoil cases converge to the residual target requested by their case files. At zero incidence the computed lift is small in every case (table 8), i.e. symmetric to within the discrete asymmetry of the triangulated mesh, while the surface pressure distributions are strongly non-trivial (C_p ranges over more than two units), so the symmetry check is not passing by triviality.

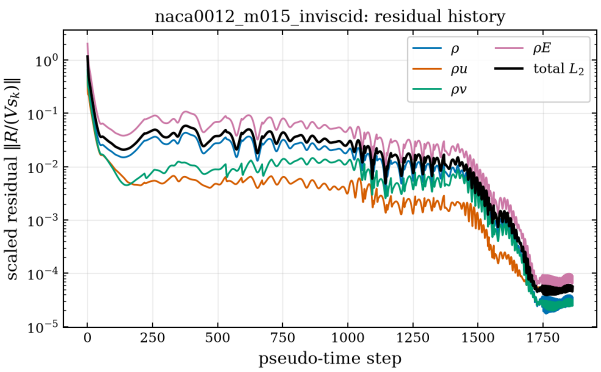
Because these are inviscid steady solutions, total enthalpy $H = \gamma p/((\gamma-1)\rho) + \frac{1}{2}|\mathbf{u}|^2$ should be uniform. It is: the area-weighted RMS of $|H - H_\infty|/H_\infty$ over the whole domain is 6×10^{-7} at $M_\infty = 0.15$ and 6×10^{-6} at $M_\infty = 0.8$ (`report/sanity_checks.json`). The pointwise maximum is much larger (5×10^{-3} and 0.10 respectively) but is confined to the sub-micron first cell layer of this viscous-type grid, whose aspect ratio reaches 2821; those cells carry a negligible fraction of the domain area. The practical consequence is that the `mach` column of `surface.csv`, which is the boundary-state Mach number of that first cell, reads slightly below the isentropic value implied by the `cp` column of the same row. The pressure — and therefore C_p and the forces — is unaffected, because pressure varies negligibly across a cell that thin.

$M_\infty = 0.15$. The flow is essentially incompressible. The computed drag $C_D = 0.00103$ is the numerical residue of d'Alembert's paradox — the exact inviscid answer is zero — and its smallness is a direct measure of the dissipation of the scheme on this mesh. Figure 2 shows the residual and force histories, the surface C_p, and the Mach and pressure fields.

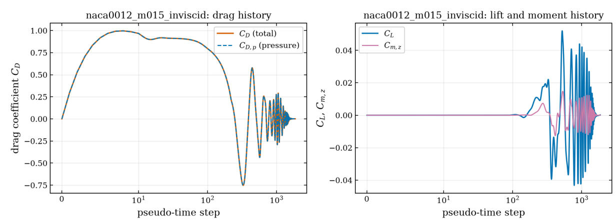
$M_\infty = 0.8$. This condition is just above the critical Mach number of the section, and the solution is transonic. The surface pressure crosses the sonic value $C_p^* = -0.435$ at $x/c \approx 0.065$, reaches $C_p = -0.932$ (an isentropic Mach number of 1.26) near mid-chord, and recovers abruptly to $C_p \approx -0.14$ between $x/c = 0.50$ and $x/c = 0.55$: a supersonic pocket about half a chord long terminated by a shock, symmetric on the two surfaces (fig. 3). The maximum Mach number in the field is 1.29. The computed $C_D = 0.00854$ is entirely wave drag — there is no viscosity and, at zero incidence, no induced drag — and is of the magnitude reported by Euler solutions of this configuration [16]. Being a near-critical condition, the pocket length and hence the wave drag are more sensitive to mesh and scheme than the subsonic case.

$M_\infty = 2.0$. A detached bow shock stands ahead of the blunt leading edge, oblique shocks and expansion fans follow the surface, and a recompression shock closes the flow at the trailing edge (fig. 4). The computed $C_D = 0.09192$ is wave drag; linearised supersonic theory for a 12%-thick section gives the same order of magnitude, and the field shows the compressible shock/expansion pattern expected at $M = 2$ rather than an incompressible-like solution. Figure 5 shows the wide view of this case beside the subsonic one, which is where the difference in character is most obvious.

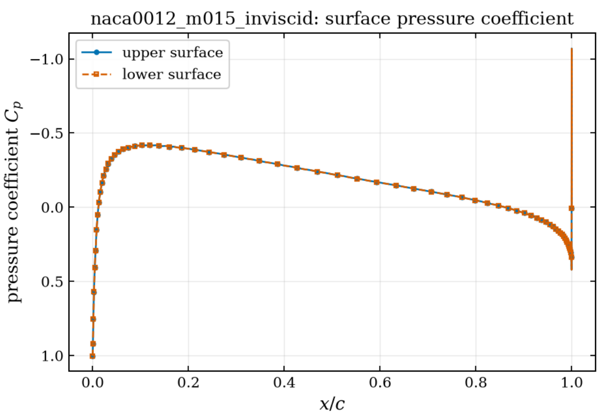
Two exact properties make this the sharpest plausibility check in the set, and both are reported rather than assumed. The section is symmetric at zero incidence, so C_L must vanish; the computed value is 8.0×10^{-4}, i.e. 0.9% of C_D. And no point on the body can carry more stagnation pressure than a normal shock at $M_\infty = 2$ delivers, which caps the surface pressure coefficient at 1.657; the computed maximum is 1.768. That the peak still exceeds the ceiling by 7% is a genuine, disclosed discretisation error: the bow shock stands off the leading edge by a fraction of the nose radius and is captured across cells whose wall-normal size is 5×10^{-6} chords, so the stagnation streamline is under-resolved. Without the shock fix of eq. (20) the same two errors are 4.6×10^{-3} and 2.120 (table 3), which is why the fix is on by default.



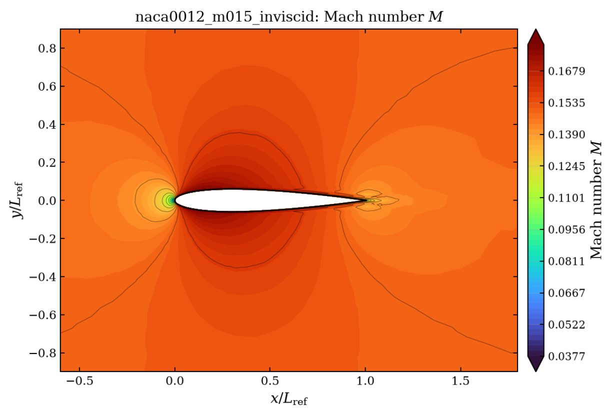
(a)



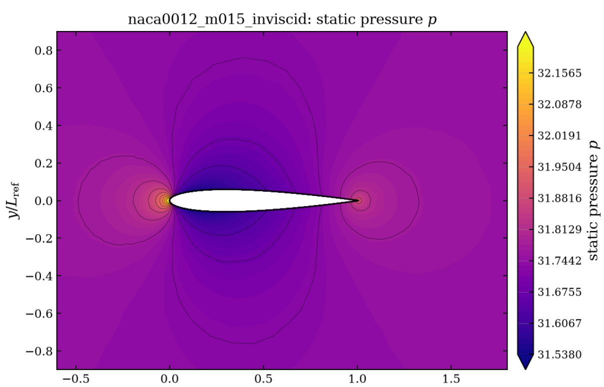
(b)



(c)



(d)



(e)

Figure 2: NACA0012, $M_\infty = 0.15$, inviscid. (a) per-equation and total residual history from `residuals.csv`; (b) drag/lift/moment history from `forces.csv`; (c) surface pressure coefficient from `surface.csv`; (d) Mach-number contours and (e) static-pressure contours from `field_final.vtu`, near-body window.

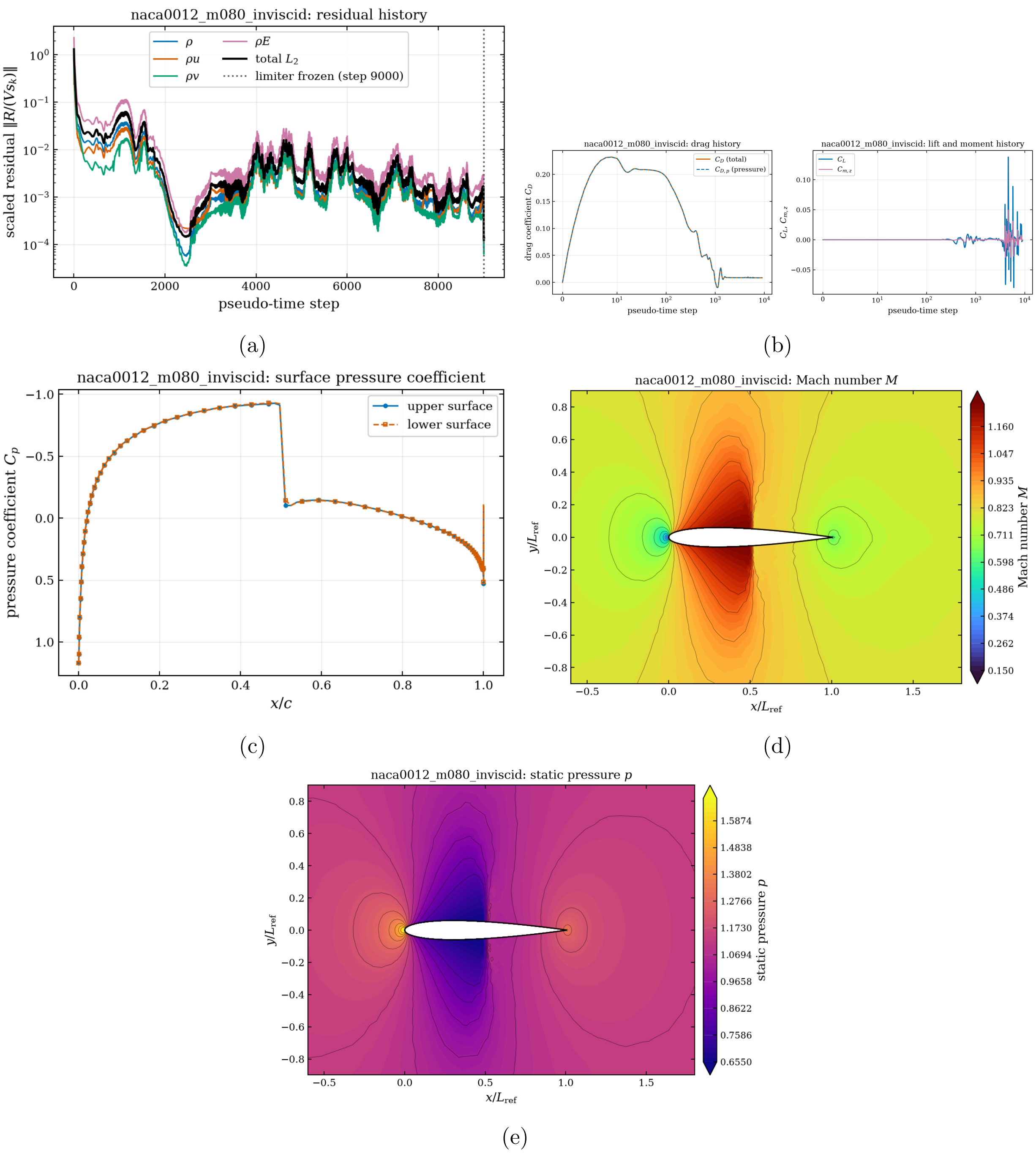


Figure 3: NACA0012, $M_\infty = 0.8$, inviscid: (a) residual history, (b) force history, (c) surface C_p showing the supercritical pocket and terminating shock, (d) Mach number and (e) static pressure near the aerofoil.

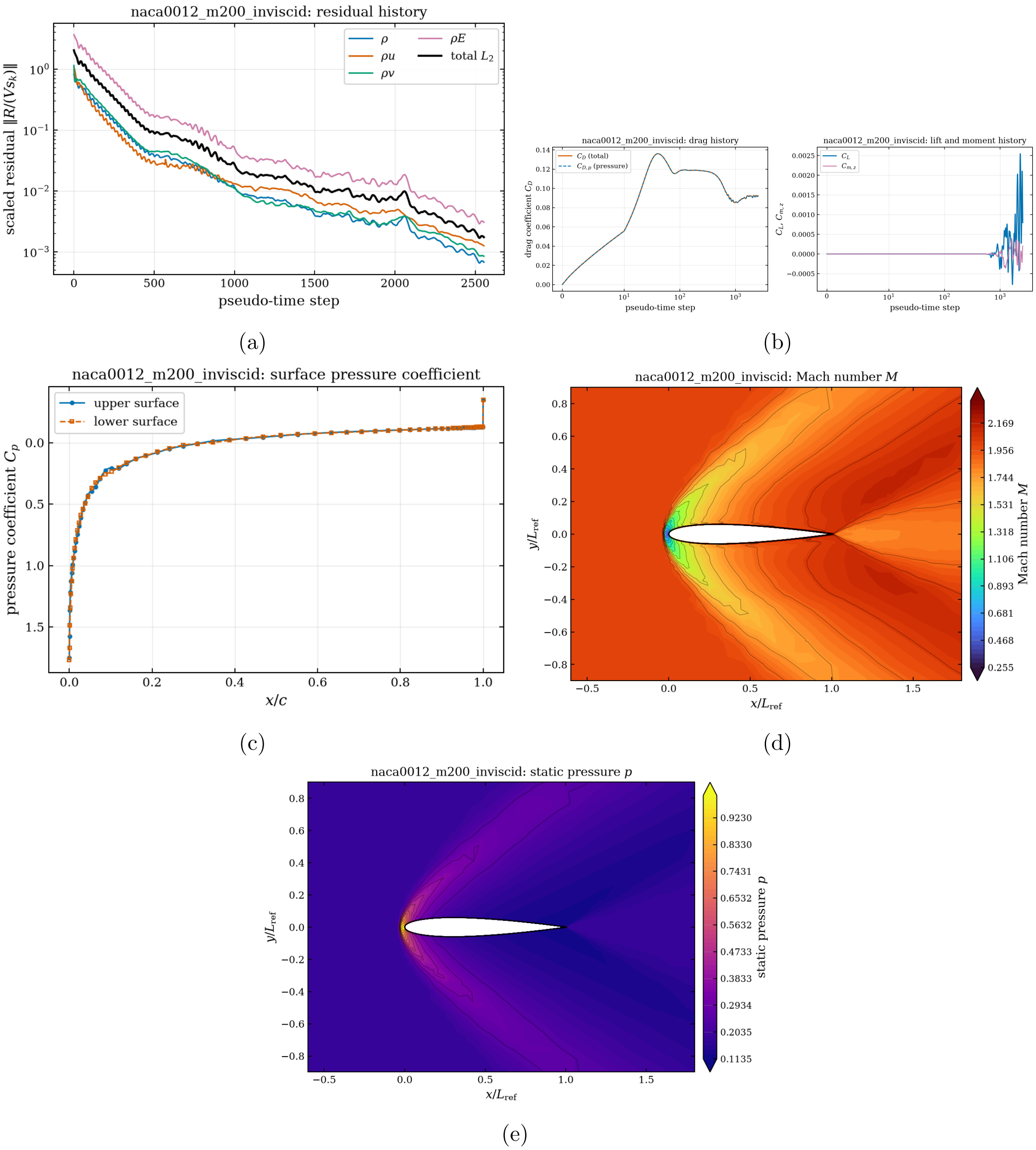
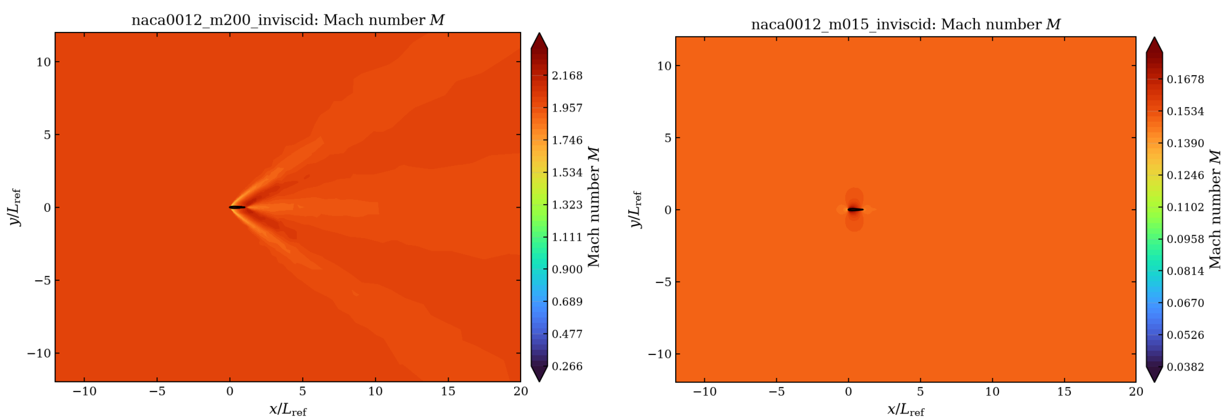


Figure 4: NACA0012, $M_\infty = 2.0$, inviscid: (a) residual history — the run reaches the requested three-order reduction while still inside the bounded limiter limit cycle, which is why the drag carries a few tenths of a percent of uncertainty (section 14); (b) force history; (c) surface C_p; (d) Mach number showing the detached bow shock and the trailing-edge recompression; (e) static pressure.



(a) (b)

Figure 5: Wide views of the Mach-number field, showing how the solution recovers towards the freestream away from the body: (a) $M_\infty = 2.0$, where the bow shock and the trailing shock system extend across the near field; (b) $M_\infty = 0.15$, where the perturbation has decayed within a few chords. The computational domain extends to 80 chords in each direction, far outside these views. Corresponding views for the other six cases are in `report/figures/*_mach_farfield.png` and are listed in the figure manifest.

10.3 NACA0012, laminar $Re = 5000$

The viscous aerofoil cases use the no-slip adiabatic wall on `bc-4` and constant viscosity from eq. (9), $\mu = 2 \times 10^{-4}$. At $Re = 5000$ the laminar boundary layer is thick and the three cases behave quite differently.

$M_\infty = 0.15$ is the cleanest check. Where the pressure gradient is weak the computed skin friction agrees with the Blasius flat-plate value to 1.9% ($C_f = 0.0169$ at $x/c = 0.3$ against $0.664/\sqrt{Re_x} = 0.0171$); it falls below it under the adverse gradient aft of maximum thickness, and crosses zero at $x/c \approx 0.90$: laminar separation ahead of the trailing edge, which is what is expected at this Reynolds number. Upper and lower surfaces agree to four digits.

$M_\infty = 0.8$ is transonic, but markedly less so than the inviscid case at the same Mach number. The thick laminar boundary layer displaces the flow and weakens the supersonic pocket: the minimum surface C_p is -0.609 against -0.932 inviscid (the sonic value is $C_p^* = -0.435$), and the peak Mach number in the field drops from 1.29 to 1.06. The boundary layer separates at $x/c \approx 0.55$, immediately downstream of where the pocket closes — a weak but unmistakable shock/boundary-layer interaction, and a genuine viscous–inviscid effect rather than a mesh artefact.

$M_\infty = 2.0$ is in the strong viscous-interaction regime. At $x/c = 0.5$ the boundary layer is $\delta_{99} = 0.062$ chords thick (`tools/mesh_facts.py`), i.e. 1.18 times the local half-thickness of the section itself, so the effective body is markedly blunter than the aerofoil. The consequence is visible in the force split: the pressure drag rises from 0.09192 in the inviscid case to 0.0971 here, on top of a skin-friction drag of 0.0413.

Reported wall rows have exactly zero velocity, as required, while the adjacent cell-centre velocities in `surface_cellcenter.csv` are finite — the two are never conflated. Figure 9 compares the vorticity fields of the three cases, which is the clearest single view of how the boundary layer and wake change with Mach number. Velocity-magnitude, wide-view and partition plots of every case are also generated and registered in the figure manifest, although they are not all reproduced here.

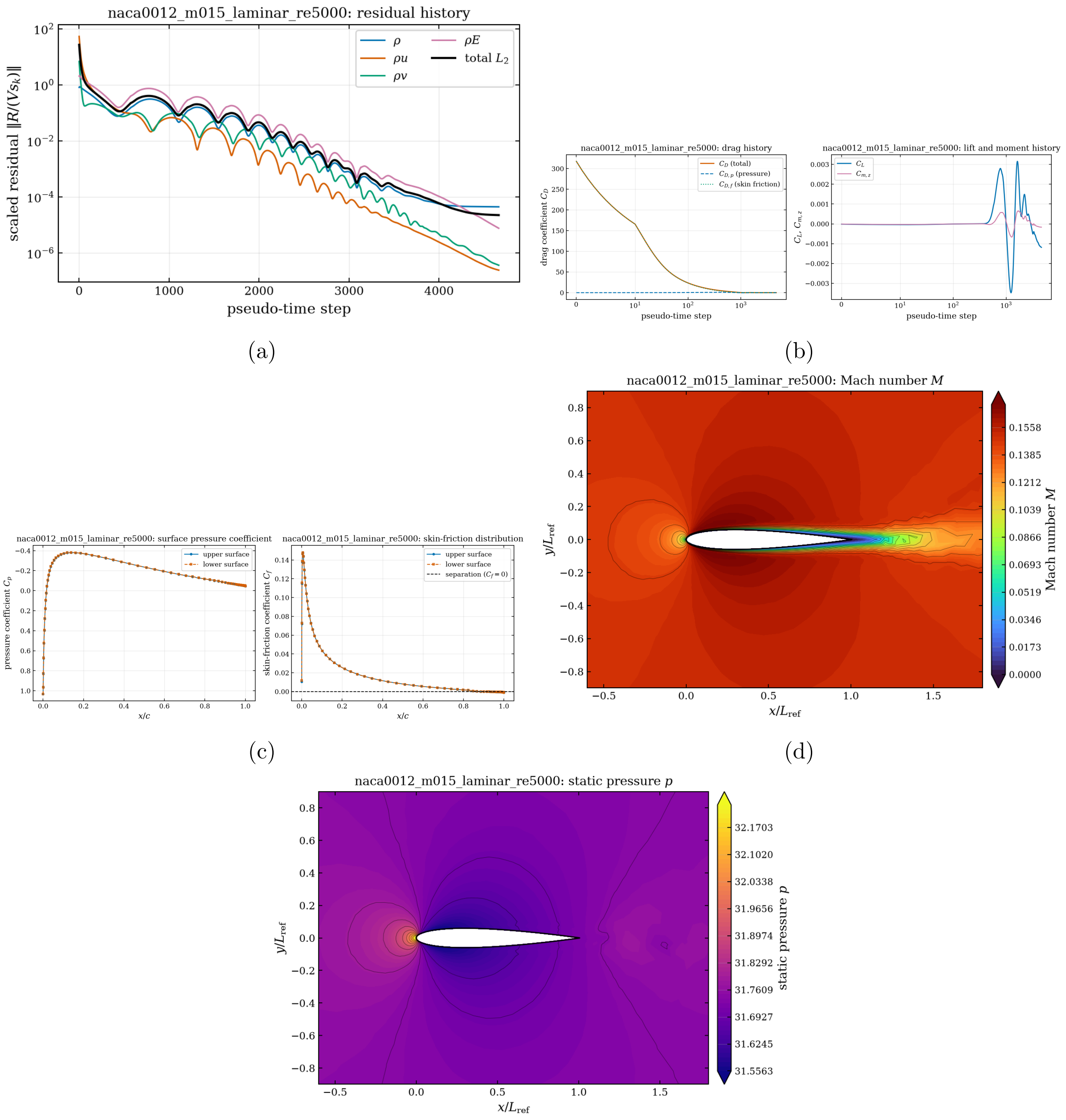
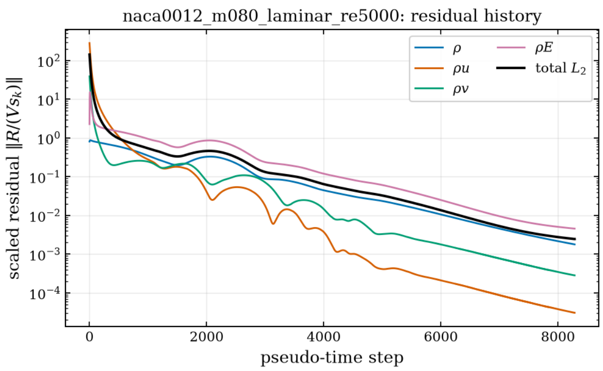
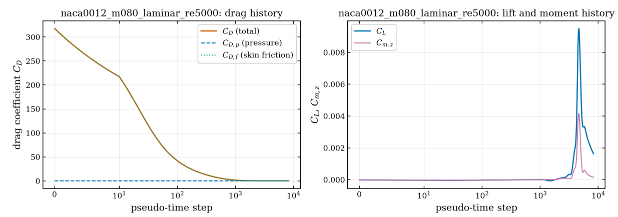


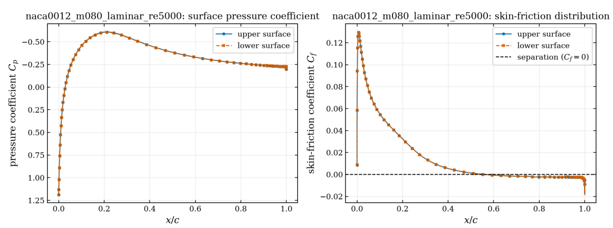
Figure 6: NACA0012, $M_\infty = 0.15$, laminar $Re = 5000$: (a) residual history, (b) force history with the pressure/skin-friction split, (c) surface C_p and skin friction C_f, (d) Mach number, (e) pressure.



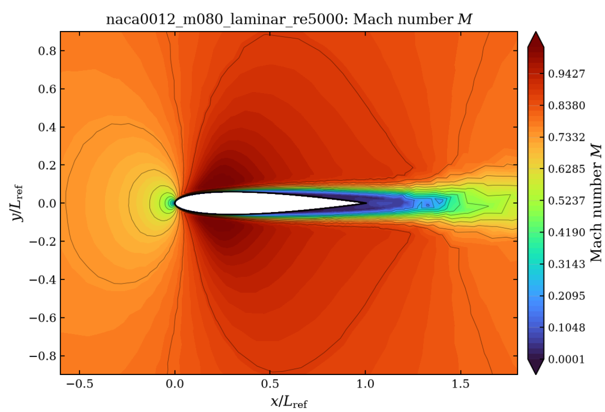
(a)



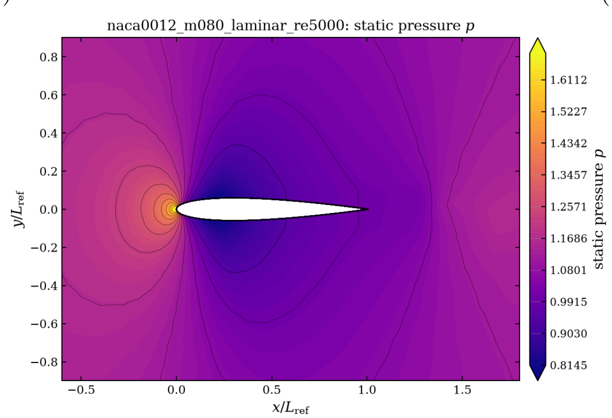
(b)



(c)



(d)



(e)

Figure 7: NACA0012, $M_\infty = 0.8$, laminar $Re = 5000$. (a) residual history, (b) force history, (c) C_p and C_f, (d) Mach number, (e) pressure. Unlike the inviscid case at the same Mach number, the surface never becomes supersonic: the thick laminar boundary layer displaces the flow so that the small supersonic region ($M_{\max} = 1.06$) sits off the wall, and the boundary layer separates at $x/c \approx 0.55$ under the resulting adverse gradient.

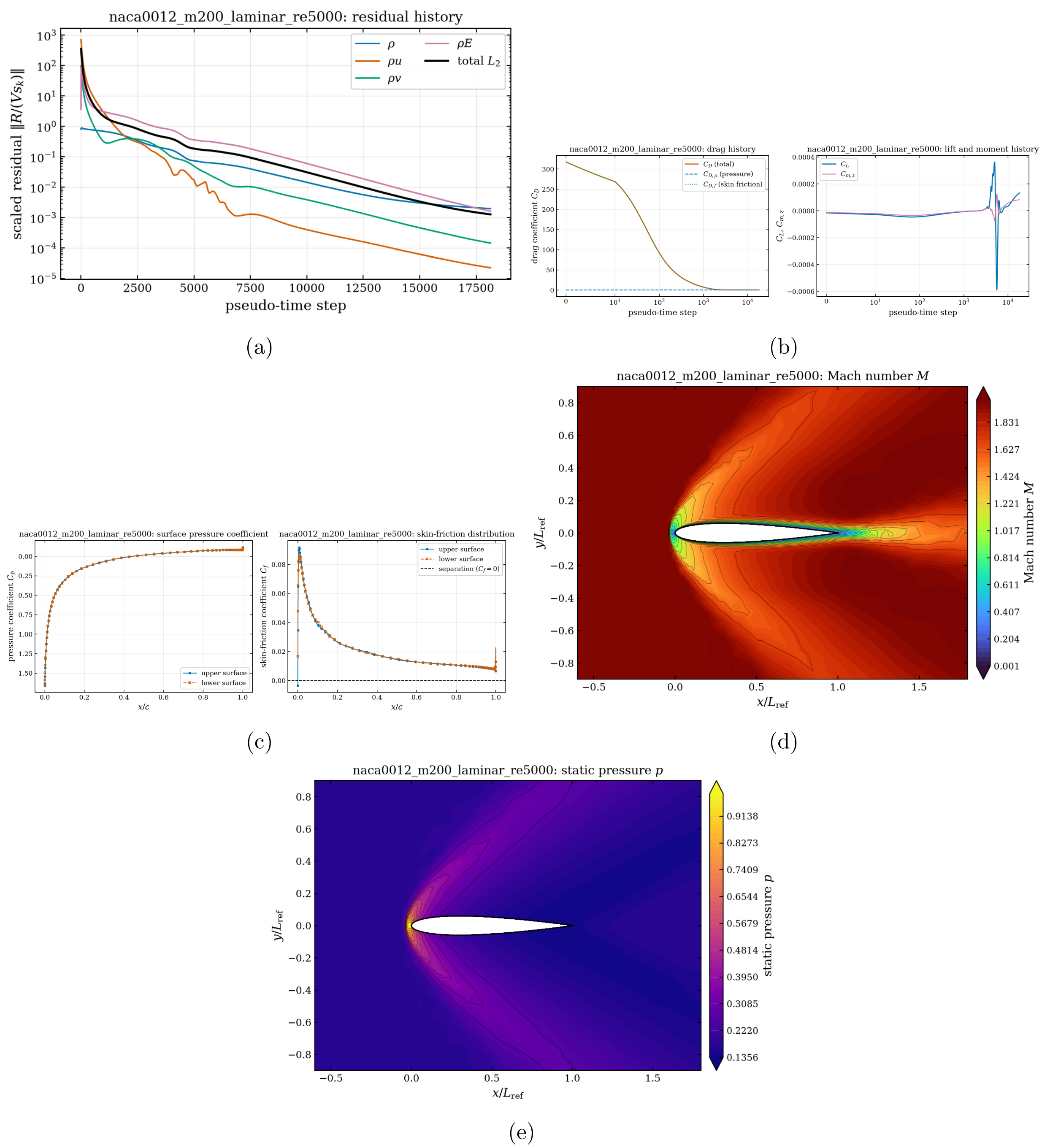
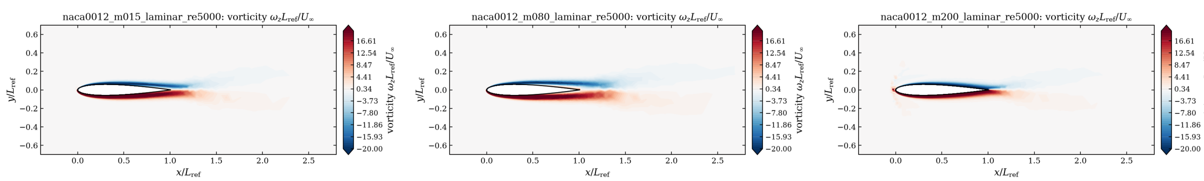


Figure 8: NACA0012, $M_\infty = 2.0$, laminar $Re = 5000$: (a) residual history, (b) force history, (c) C_p and C_f, (d) Mach number, (e) pressure.



(a) (b) (c)

Figure 9: Vorticity $\omega_z L_{\mathrm{ref}}/U_\infty$ of the three laminar aerofoil cases, clipped to $[-20, 20]$: (a) $M_\infty = 0.15$, (b) $M_\infty = 0.8$, (c) $M_\infty = 2.0$. The two boundary layers of opposite sign leave the trailing edge and merge into the wake; the wake is thickest at $M_\infty = 0.8$, where the shock separates the boundary layer at mid-chord, and thinnest at $M_\infty = 2.0$. Source: `field_final.vtu` of each case.

10.4 Circular cylinder, $Re = 20$

This is the cleanest quantitative validation in the set, because the low-Reynolds cylinder is a classical reference problem with well-established reference values [12, 13]. Every quantity below is read from `results/cylinder_m010_laminar_re20/surface.csv` and `forces.csv`.

Table 11: Circular cylinder at $Re = 20$: computed quantities against accepted values from the steady low-Reynolds cylinder literature [12, 13]. Every computed number is read from `results/cylinder_m010_laminar_re20/surface.csv` and `forces.csv` by `tools/make_report_tables.py`.

quantity	computed	accepted
total drag C_D	2.0176	2.0–2.05
pressure drag $C_{D,p}$	1.2198	≈ 1.23
skin-friction drag $C_{D,f}$	0.7978	≈ 0.81
lift C_L (symmetry)	+0.000643	0
front stagnation C_p	1.2683	≈ 1.26
base pressure C_p at $\theta = 0°$	-0.5557	≈ -0.55
minimum C_p	-0.9933 at 99° from stagnation	–
separation angle from the rear	41°–45°	43°–45°

The agreement is within 2% on the total drag and on both parts of its split, and the separation angle falls in the accepted range. Two details are worth noting. First, the front stagnation C_p sits well above the compressible stagnation value of 1.002 at $M_\infty = 0.1$; that excess is the expected low-Reynolds viscous enhancement of the stagnation pressure and is not a numerical artefact. Second, the skin friction changes sign a little over 40° ahead of the rear stagnation line, which is where the steady recirculation bubble begins; the sign convention of section 5 makes that crossing directly readable from the C_f plot. A steady, closed, symmetric wake is formed (fig. 10).

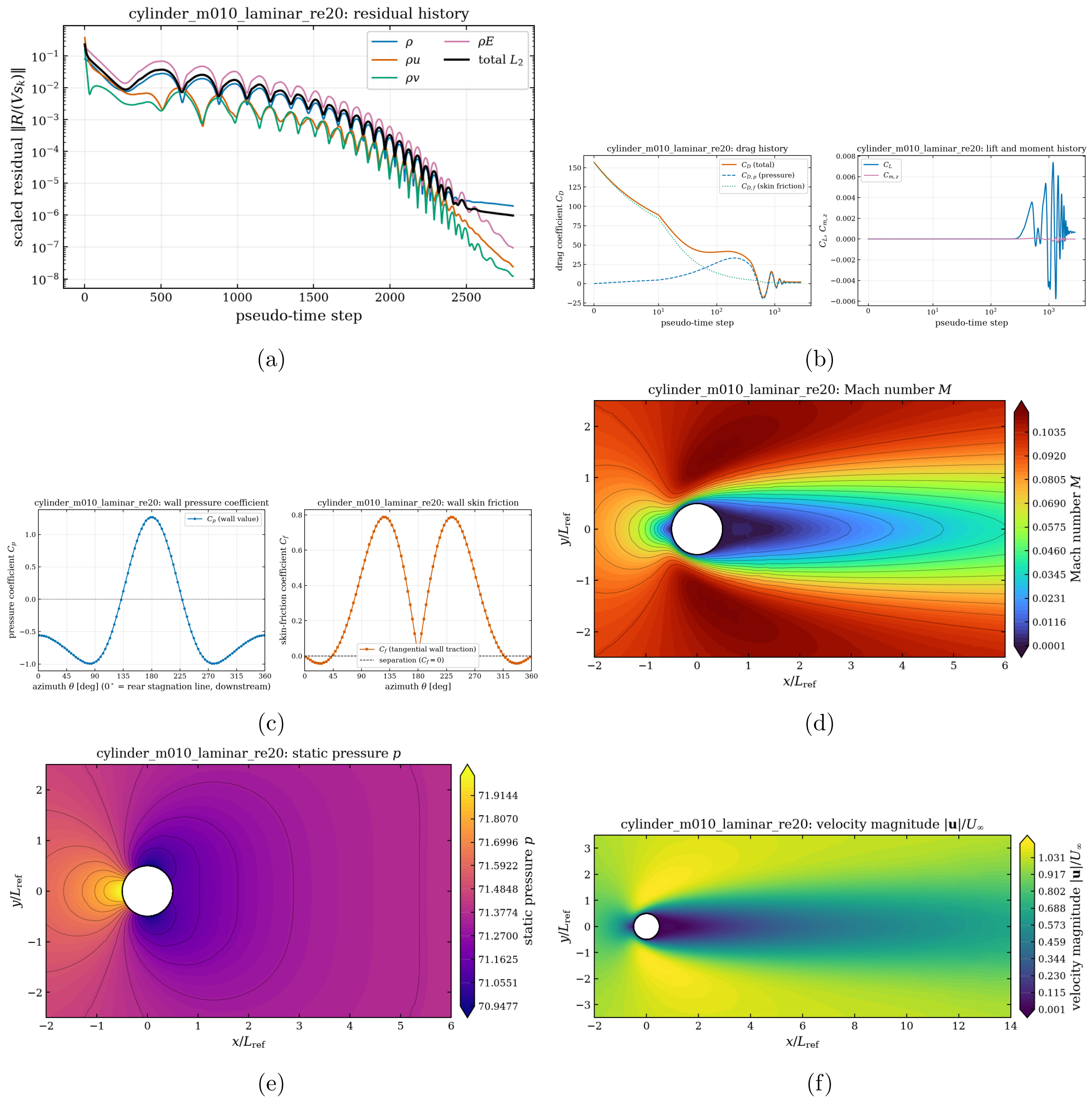


Figure 10: Cylinder, $Re = 20$: (a) residual history, (b) force history with the pressure/friction split, (c) wall C_p and C_f versus azimuth ($\theta = 0$ is the downstream stagnation line), (d) Mach number, (e) pressure, (f) velocity magnitude showing the steady closed wake.

10.5 Circular cylinder, $Re = 200$: the vortex street

The transient case is run with the supplied production controls ($\Delta t = 0.01$ to $t_f = 300$, i.e. 30000 physical steps, inner iterations bounded by 5 and 1000, BDF2 with frozen histories) and the tightened dual-time settings justified in section 6. The flow starts impulsively, the initially symmetric recirculation becomes unstable, and a periodic von Kármán street is established well before the end of the record.

Analysis over the second half of the record ($t \in [150, 300]$, performed by `tools/analyze_transient.py` and stored in `report/shedding_analysis.json`) gives a shedding frequency of 0.1859 (spectral peak) and 0.1857 (mean of 27 lift up-crossing intervals), i.e. a Strouhal number $St = fD/U_\infty = 0.1859$, against the accepted experimental and numerical value of 0.19–0.20 at $Re = 200$ [14, 15]. The mean drag is $C_D = 1.2661$ (accepted 1.32–1.40 [15, 14]), split into $C_{D,p} = 1.0243$ and $C_{D,f} = 0.2417$, and the lift oscillates with amplitude 0.5745 (RMS 0.4069) against an accepted amplitude of 0.6–0.7. The drag oscillates at twice the lift frequency with amplitude 0.0280, as it must for a symmetric body.

Table 12: Circular cylinder at $Re = 200$: computed post-transient quantities, averaged over $t \in [150, 300]$, against accepted values for the two-dimensional laminar vortex street [14, 15]. Source: `report/shedding_analysis.json`, computed from `results/cylinder_m010_laminar_re200/forces.csv`.

quantity	computed	accepted
Strouhal number $St = fD/U_\infty$ (spectral peak)	0.1859	0.19–0.20
Strouhal number (lift up-crossings)	0.1857	0.19–0.20
mean drag $\overline{C_D}$	1.2661	1.32–1.40
pressure part	1.0243	–
skin-friction part	0.2417	–
lift amplitude	0.5745	0.60–0.69
lift RMS	0.4069	0.42–0.49
drag oscillation amplitude	0.0280	–
mean lift (symmetry)	+0.00173	0
shedding cycles analysed	27	–

Table 12 collects the comparison. All three quantities sit slightly *below* the accepted ranges, and consistently so: the Strouhal number by 2.2%, the mean drag by 4.1% and the lift amplitude by 4.3%, each measured against the lower edge of its band. A single systematic shortfall in all three is more informative than three separate discrepancies would be, because the same mechanism — a wake that is slightly weaker than it should be — lowers all of them together. Three effects contribute in that direction and none of them is a defect of the scheme: the reference values are for incompressible flow while these computations are at $M_\infty = 0.1$, which lowers the drag slightly; the upwind dissipation of HLLC is not low-Mach preconditioned (section 14), so it is larger than it would be at the same cell size in an incompressible solver; and the supplied mesh coarsens quickly downstream, so the shed vortices lose strength as they convect. The deficit is reported rather than tuned away.

The post-transient force records and the lift spectrum used for this analysis are shown in fig. 11; the wake structure at the final time is shown in fig. 12, and the complete history from the impulsive start, together with the per-step inner-iteration count and the instantaneous wall distributions, in fig. 13.

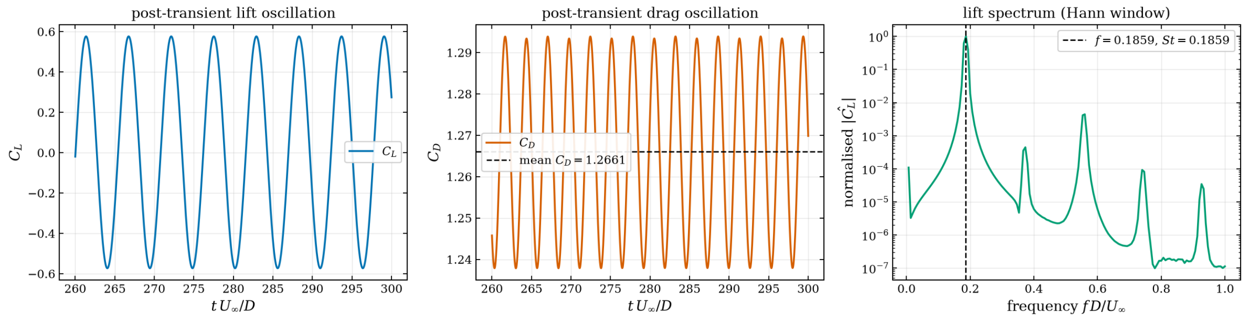
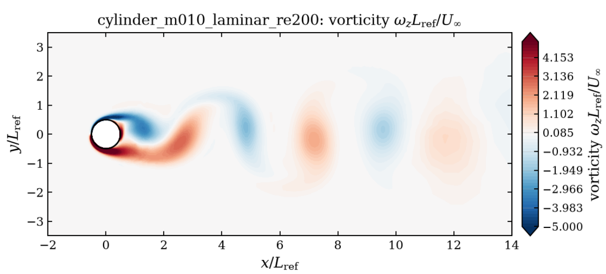
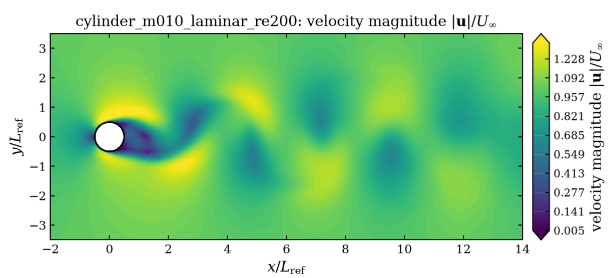


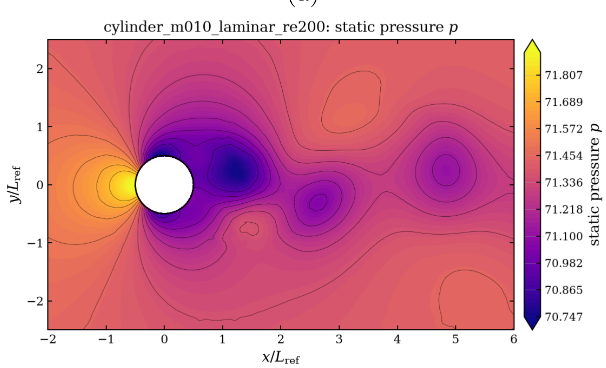
Figure 11: Reynolds 200 cylinder: post-transient lift and drag oscillations and the Hann-windowed lift spectrum used to extract the shedding frequency. Source: `results/cylinder_m010_laminar_re200/forces.csv`.



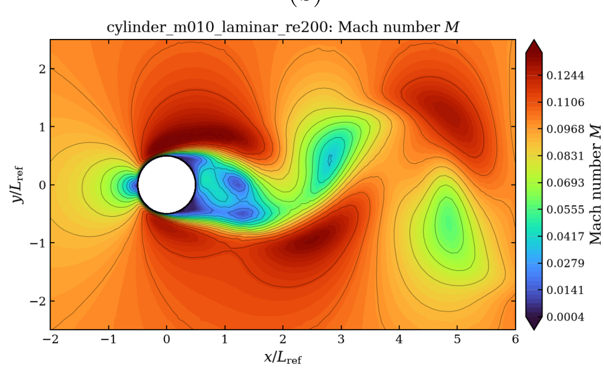
(a)



(b)



(c)



(d)

Figure 12: Reynolds 200 cylinder at the final time $t = 300$: (a) vorticity with the recommended clipped range $[-5, 5]$, showing the alternating von Kármán street; (b) velocity magnitude over the same wake window; (c) static pressure with the low-pressure vortex cores; (d) Mach number. All from `field_final.vtu`.

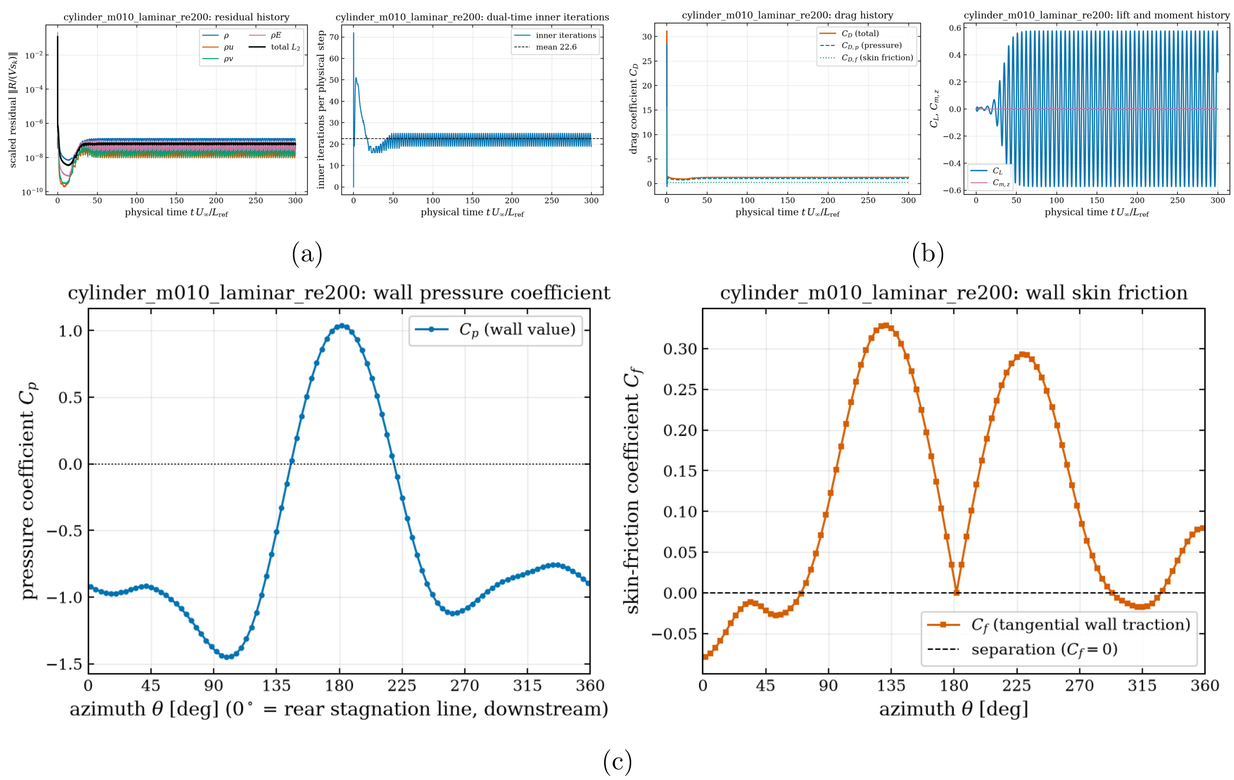


Figure 13: Reynolds 200 cylinder: (a) total dual-time residual per physical step and the number of inner iterations used at each step; (b) the complete force history from the impulsive start to $t = 300$; (c) instantaneous wall pressure coefficient and skin friction at the final time $t = 300$, which are asymmetric about $\theta = 180°$ because the vortex street is shedding.

11 Parallel Validation

Table 13 and fig. 14 compare one NACA case and one cylinder case across $np = 1, 2, 4, 8$. Every production run in table 7 was executed on 8 ranks, so the $np = 8$ requirement is satisfied for both geometries by the submitted results themselves; the rank study is an additional set of runs stored in `studies/mpi/`.

Force coefficients agree closely across rank counts. For the cylinder ($C_D = 2.0176$) the relative difference against the single-rank run is at most 3.9×10^{-6}. For the aerofoil the relative figure reaches 1.1×10^{-2}, but that is a relative measure of a drag coefficient of 0.0010 — the spurious drag of a symmetric, uncambered section at zero incidence in an inviscid flow, whose exact value is zero — so the meaningful measure is the *absolute* disagreement, 1.1×10^{-5}. Both are at the level of the iterative convergence criterion — the runs stop when the mean C_D moves by less than $\max(5 \times 10^{-5}, 3 \times 10^{-3}|C_D|)$ between two consecutive windows, which is 5×10^{-5} for this aerofoil case and 6.1×10^{-3} for the cylinder — rather than a partition effect: the residual assembly is partition-independent by construction (section 7), and the only rank-dependence is the lagged coupling of the LU-SGS sweeps across rank boundaries, which changes the path to the steady state but not the state itself. The step counts in table 13 show exactly that — they differ by a few percent between rank counts while the converged forces do not. Load balance stays within 1.5% of perfect and the METIS edge cut grows roughly linearly with the rank count, as expected for a two-dimensional partition.

Table 13: MPI rank-count study. Speed-up is relative to the single-rank run of the same case. $\Delta C_D/C_D$ is measured against the single-rank result.

case	ranks	steps	wall [s]	speed-up	C_L	C_D	$\Delta C_D/C_D$	edge cut	balance
cylinder_m010_laminar_re20	1	2803	42.4	1.00	+0.000648	2.017610	0	0	1.0000
cylinder_m010_laminar_re20	2	2802	20.9	2.03	+0.000647	2.017616	3.24×10^{-6}	97	1.0003
cylinder_m010_laminar_re20	4	2805	11.6	3.65	+0.000653	2.017618	3.91×10^{-6}	282	1.0007
cylinder_m010_laminar_re20	8	2806	18.1	2.35	+0.000643	2.017617	3.51×10^{-6}	471	1.0070
naca0012_m015_inviscid	1	1906	52.3	1.00	+0.000183	0.001014	0	0	1.0000
naca0012_m015_inviscid	2	1844	26.9	1.94	+0.000197	0.001023	8.99×10^{-3}	120	1.0001
naca0012_m015_inviscid	4	1894	15.2	3.44	+0.000066	0.001021	6.72×10^{-3}	241	1.0019
naca0012_m015_inviscid	8	1864	24.0	2.18	+0.000147	0.001025	1.11×10^{-2}	439	1.0154

Table 14: Per-rank partition diagnostics of the 8-rank `naca0012_m015_inviscid` production run (`partition_diagnostics.csv`).

rank	owned	ghost	boundary faces	neighbours	sent	received
0	2626	136	47	6	136	136
1	2631	93	66	2	93	93
2	2593	104	58	3	105	104
3	2552	87	76	2	86	87
4	2578	99	63	2	99	99
5	2578	102	64	4	102	102
6	2642	62	80	2	62	62
7	2616	188	30	5	188	188
total	20816	871	484		871	871

The partitions actually used by the two 8-rank runs are shown in fig. 15.

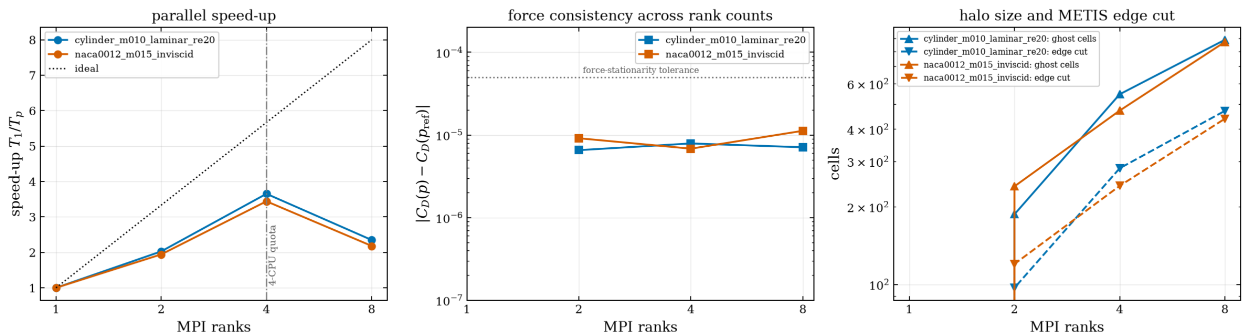
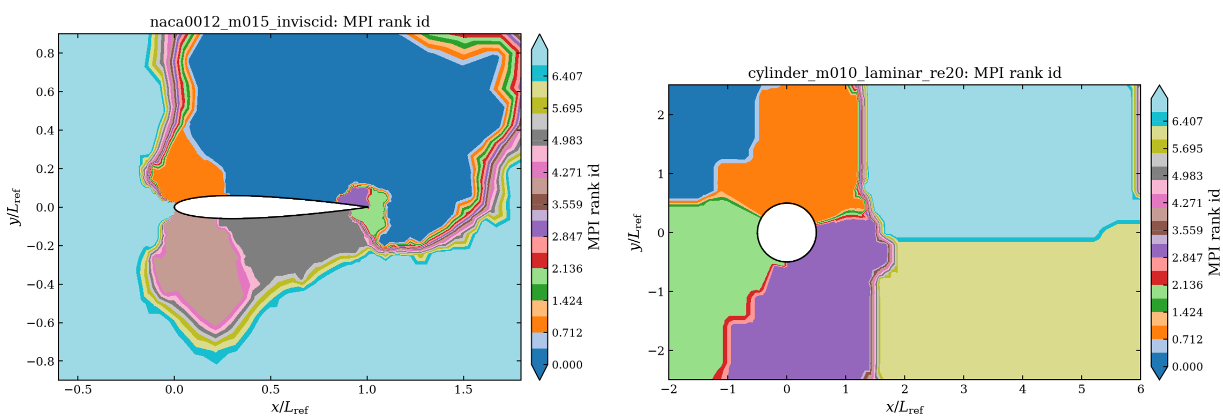


Figure 14: Rank-count study: parallel speed-up, force consistency and halo growth. In the left panel the dash-dotted line marks the 4-CPU cgroup quota of the benchmark container; beyond it the 8-rank runs place two ranks on every available CPU, which is why the speed-up collapses. The centre panel shows the *absolute* difference in C_D from the single-rank run against the absolute force-stationarity tolerance at which the runs were stopped (dotted); every rank count agrees with $np = 1$ well inside it. Source: `report/mpi_study.csv`.



(a) (b)

Figure 15: METIS k-way partitions actually used by the 8-rank runs, coloured by the rank that owned each cell during the iterations (field `RankId` of `field_final.vtu`).

Timing caveat. The benchmark container advertises 64 logical cores but is limited to a 4-CPU cgroup quota (`cpu.max` = 400000/100000). Speed-up is therefore meaningful only up to $np = 4$, where it reaches 3.65× (cylinder) and 3.44× (aerofoil) — 91% and 86% of the four CPUs the container is actually allowed. Beyond that there is nothing left to win: the $np = 8$ runs place two ranks on every available CPU and fall back to 2.35× and 2.18× while performing the same work and producing the same answer. Open MPI is also launched with `--bind-to none`, because its default core pinning collapses performance under such a quota; this was measured during development and is the reason `scripts/env.sh` sets `MPIRUN_FLAGS` as it does. The host is shared, so the wall-clock column of table 13 carries run-to-run noise of the order of ten percent and should not be read as a clean scaling curve; the rank-independent columns (steps, forces, edge cut, balance) are the ones that test the solver. All of this is a property of the execution environment, not of the parallelisation, and is stated here rather than hidden. The $np = 8$ requirement is a correctness demonstration, and it is met: every submitted production result in table 7 was computed on 8 ranks.

12 Code Organisation and Extensibility

The solver is 6 518 lines of C++ split into seven modules with one responsibility each: `core/` (types, case-file parsing and validation, logging), `mesh/` (CGNS import, global mesh, METIS partitioner, distributor, rank-local mesh), `parallel/` (halo exchange and reductions), `physics/` (equation of state, transport, Riemann solvers, viscous flux, boundary conditions), `numerics/` (gradients, limiters, residual assembly, LU-SGS, the steady and transient drivers, verification), `io/` and `post/`. No file exceeds 660 lines. The Python post-processing is a further 2 716 lines under `tools/`, none of which participates in the solve. Both counts are produced by `tools/make_report_tables.py` from the tree itself.

The seams that matter for the extensions the task asks about are:

Dimension and variable count. `kDim` and `kNVar` are the only place where "two-dimensional, four equations" is written down (`src/core/Types.hpp`). The partitioner, distributor, halo exchange, least-squares stencils, limiter and implicit solver are all written against them and against a general polygonal cell, not against a fixed cell type. What a 3-D extension genuinely requires is a polyhedral face/cell topology in `GlobalMesh`/`LocalMesh` and the 3-D CGNS element mapping; the numerics interfaces above them do not change. That is a real piece of work, not a recompile, and the report does not claim otherwise.

Equation of state. Everything thermodynamic goes through the small `PerfectGas` interface (`pressure`, `soundSpeed`, `temperature`, `toPrimitive`, `toConservative`, `normalFlux`, `totalEnthalpy`). A real-gas or Cantera-backed model is a sibling class with the same members; Cantera is already available in the environment and linked-in support would not touch the residual assembly.

Boundary conditions. Each type is two pure functions — the exterior state for the Riemann solver and the physical boundary value — selected by an enumerator parsed from the case file. Adding an isothermal wall, a subsonic inlet or a pressure outlet is one enumerator plus one branch in each function, with no change anywhere else.

Transport and turbulence. `TransportModel` already carries constant and Sutherland laws behind a single `viscosity(T)` call. A RANS eddy viscosity enters at exactly that point plus one extra transported variable, which is what `kNVar` controls.

Fluxes and limiters. The Riemann solver and the limiter are selected by enumerator at run time and are pure functions of their arguments (`FluxInviscid.hpp`, `Limiter.hpp`), which is why the flux cross-check of table 6 and the limiter comparison of table 5 are single command-line switches rather than rebuilds.

Nothing in `src/` branches on a case identifier, a mesh file name or a boundary-family name: a repository-wide search for the eight case ids, the two mesh file names and the four family names of the supplied meshes returns matches only in comments and in the `case_id` metadata label. Case differences come exclusively from the JSON inputs and from the documented command-line options listed in table 9.

13 Visualisation and Plot-Style Conventions

All figures are produced by `tools/make_figures.py` from the submitted output files and share one style module (`tools/plot_style.py`).

1. **Line plots** use a serif 11 pt font, a colour-blind-safe qualitative palette, line widths of 1.1–1.8 pt, labelled axes with nondimensional quantities, a legend on every panel, a light grid and inward ticks on all four sides. Residual histories are logarithmic in the ordinate; steady force histories use a symmetric-log abscissa so the startup transient and the converged plateau are both legible.

2. **Field plots** are filled contours (`tricontourf`) evaluated on the *actual* mesh: every cell is split into triangles from its own connectivity, and cell values are moved to the nodes by area-weighted averaging. No interpolation grid and no scatter plot is used anywhere. Every filled contour plot uses 60 levels, and the near-body Mach and pressure views additionally carry 12 thin contour lines so that the shape of the isolines is readable and not only the colour. The line levels exclude the outer 15% of the plotted range: the outermost bands are the undisturbed freestream and the no-slip wall, large nearly uniform plateaus whose ragged edge a contour line would trace without conveying anything.

3. **Colour bars** are always present and always carry the variable name and its nondimensional definition. Ranges of the Mach, pressure and velocity-magnitude plots are clipped to the 1–99 percentile *of the plotted window*, so that a handful of cells inside a shock or at a stagnation point cannot collapse the range for the rest of the field; the clip is recorded in the caption column of `report/figure_manifest.csv` for every affected figure. Vorticity uses a fixed symmetric range on a diverging colour map centred on zero: $[-5, 5]$ for the cylinder cases, which is the range recommended in the case file, and $[-20, 20]$ for the aerofoil cases, whose boundary-layer vorticity is an order of magnitude larger.

4. **View windows** are chosen per geometry: a near-body window for the aerofoil ($-0.6 \leq x \leq 1.8$), a wake window for the cylinder ($-2 \leq x \leq 14$), and a wide view for the farfield recovery. The wall outline taken from `surface.csv` is overlaid on every field plot so the body is unambiguous.

5. **File naming** follows `<case_id>_<variable>.png` and each file is registered in `report/figure_manifest.csv` together with the variable it plots and the file it was made from. A `*_mach.png` plots Mach number and a `*_pressure.png` plots static pressure; the mapping is checked by `tools/sanity_checks.py` and by the benchmark validator.

14 Limitations and Failure Analysis

Residual carbuncle at the Mach 2 stagnation point. HLLC is not immune to the same class of instability, only less prone to it, and the shock fix of eq. (20) suppresses rather than eliminates it. After the fix the Mach 2 leading-edge pressure coefficient is 1.768 against the Rayleigh–Pitot ceiling of 1.657 — an excess of 7%, down from 28% without it — and the upper/lower surface C_p distributions still differ by 3.1×10^{-2} rms where the exact difference is zero. Part of that residual asymmetry is not the scheme's fault: the supplied mesh is not exactly mirror-symmetric at the leading edge. Within $|x| < 0.02c$ it carries 1206 cells above the axis against 1222 below it, and 38% of the upper cells have no mirror closer than a tenth of their own size (`tools/mesh_facts.py`), so a perfectly symmetric scheme run on this mesh would still produce a nonzero C_L. The remainder is under-resolution of the stand-off region. A pure Rusanov run, which cannot carbuncle at all, gives 1.645 and 3.7×10^{-3} rms (table 3). Two independent stabilisations agreeing is the strongest evidence available here that the fixed answer is the right one: Rusanov's drag, 0.09202, and the fixed HLLC value, 0.09192, differ by 0.1%, while the unfixed run sits 4.4% away at 0.08789. The carbuncle also costs convergence: the unfixed case needs 6.0 times as many pseudo-time steps to satisfy the same criterion. The proper cure would be a mesh with a resolved nose radius, or a rotated-hybrid Riemann solver in which the shock-normal direction is detected per face rather than blended for.

Residual limit cycle on the shock cases. As described in section 6, the non-differentiable min / max stencil of the limiter drives the shock-containing cases into a bounded residual limit cycle. Freezing the limiter removes it at Mach 0.8 (fig. 16): the frozen run reaches the requested four orders within a few steps of the freeze point, whereas the unfrozen run stalls at 1.4×10^{-3} of its initial residual for the whole 20000-step budget and its drag is still drifting upwards when that budget runs out. The two differ by 1.76% in C_D (0.008541 frozen against 0.008692 unfrozen), which is a direct measure of how much the limit cycle biases the unconverged answer; the frozen value is the converged one and is what table 8 reports. At Mach 2 the run reaches the requested three-order reduction, and the stationarity test accepts it, while the solution is still inside the cycle: over the acceptance window the drag scatter is 0.44% of C_D and the window-to-window drift is 0.30%. Continuing the same case without limiter freezing to the full 2554-step budget (`studies/verify/naca0012_m200_inviscid_nofreeze`) leaves C_D at 0.09192 against 0.09192 for the accepted production run — a 0.00% difference, inside that band. A few tenths of a percent — not the five digits printed in table 8 — is therefore the honest uncertainty of the Mach 2 drag on this mesh, and it is the reason the force-stationarity tolerance is 0.3% of $|C_D|$ rather than tighter. A differentiable limiter (an MLP-$u2$ variant or a smooth pressure-based sensor) would be the proper fix.

Limiter accuracy in smooth regions. The manufactured-solution study measures 2.03 with the limiter active against 2.30 without it, so at these resolutions the Venkatakrishnan limiter is not costing accuracy in smooth flow. That is not guaranteed to hold under further refinement: $\epsilon^2 = (Kh_c)^3 s_v^2$ vanishes faster than $\Delta^{\pm 2} \sim h^2$, so on a sufficiently fine mesh the limiter degenerates towards Barth–Jespersen and the formal order drops. Raising K postpones this at the cost of shock robustness; $K = 5$ is the compromise used throughout, and no attempt was made to tune it per case.

Low-Mach dissipation. No low-Mach preconditioning is implemented. At $M_\infty = 0.1$–0.15 the upwind dissipation of HLLC scales with the acoustic speed rather than the convective speed, which

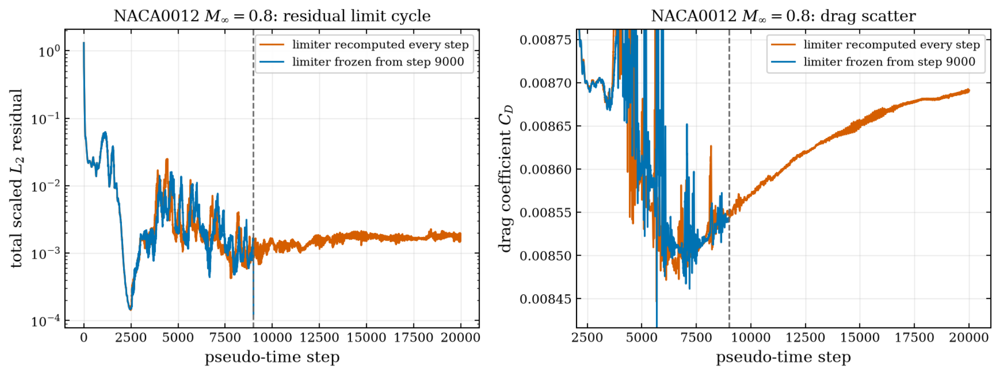


Figure 16: Mach 0.8 inviscid aerofoil: residual and drag history with the limiter recomputed at every step (bounded limit cycle, residual stalling around 10^{-3} of its initial value) and with the limiter frozen from step 9000, after which the residual reaches the requested four-order reduction within a few steps. Source: `studies/verify/naca0012_m080_inviscid_nofreeze/` and `results/naca0012_m080_inviscid/`.

adds artificial dissipation. The $Re = 20$ cylinder drag nevertheless lands within 1% of the accepted value, so the effect is tolerable on these meshes, but a Weiss–Smith-type preconditioner would improve the low-Mach cases.

Outer-field resolution of the supersonic cases. The supplied aerofoil mesh is a viscous-type grid: it puts its cells into the boundary layer and coarsens quickly away from the body, reaching cell sizes of order $0.1c$ within one chord. That is the right choice for the $Re = 5000$ cases and the wrong one for the Mach-wave field of a supersonic flow. The consequence is visible in the near-body Mach plots of the two Mach 2 cases (figs. 4 and 8): the waves radiated by the captured leading-edge shock are resolved across one or two cells and appear as rays aligned with the mesh, and they are sharper above the section than below it because the triangulation is not mirror-symmetric there. Plotting the raw cell values with flat shading gives the same picture, so this is the solution and not the contouring. The surface quantities and the integrated forces are not affected — the upper and lower surface C_p distributions of the Mach 2 laminar case agree to 1.0×10^{-2} rms and its $|C_L|$ is 1.3×10^{-4} — but a study that cared about the wave field would need a mesh adapted to it.

Single mesh per geometry. The benchmark supplies one mesh per geometry, so no grid-convergence study of the benchmark results themselves is possible; the order verification in section 8 is performed on generated meshes instead. Reported force coefficients therefore carry an unquantified spatial discretisation error.

Hybrid LU-SGS across ranks. The Gauss–Seidel coupling is exact within a rank and lagged between ranks, so the number of nonlinear iterations grows slightly with rank count. This is visible in table 13 as a small difference in the step count to reach the same residual target.

Two-dimensional only. `kDim` and `kNVar` are compile-time constants and the mesh layer is written for polygonal cells; extending to 3-D requires a polyhedral face/cell topology in `GlobalMesh`/`LocalMesh` and a 3-D CGNS element mapping, but not changes to the numerics interfaces.

15 Conclusions

`cfd2d` implements the required physics and numerics — conservative finite volume on unstructured mixed meshes, second-order limited reconstruction, approximate Riemann solvers, consistent viscous discretisation, implicit pseudo-time marching and true dual-time BDF2 — with METIS partitioning and neighbour-scoped MPI halo exchange, and it produces complete, converged or statistically periodic results for all 8 required benchmark cases with 0 failure. Independent verification gives an observed discretisation order of 2.30 for the unlimited second-order scheme and machine-level freestream preservation and linear exactness on the benchmark mesh; the two quantitative validation targets available in the literature — $C_D \approx 2.0$–2.05 for the $Re = 20$ cylinder [12, 13] and $St \approx 0.19$–0.20, $C_D \approx 1.32$–1.40 for the $Re = 200$ vortex street [14, 15] — are reproduced to within a few percent: the $Re = 20$ drag and its pressure/friction split to 2%, and the $Re = 200$ Strouhal number, mean drag and lift amplitude to 2.2%, 4.1% and 4.3% below the lower edge of their accepted bands — a single consistent shortfall rather than three independent errors, for the reasons set out in section 10.5.

The principal weaknesses, all documented in section 14, are the limiter-induced residual limit cycle on the shock-containing cases, which leaves a few tenths of a percent of uncertainty in the Mach 2 drag; a residual shock-capturing overshoot at the Mach 2 stagnation point, which the shock fix reduces to 7% of the exact bound but does not remove; the absence of low-Mach preconditioning; the coarseness of the supplied viscous mesh in the outer supersonic wave field; and the fact that the supplied single mesh per geometry makes a grid-convergence study of the benchmark results themselves impossible, so the reported coefficients carry an unquantified spatial discretisation error on top of the iterative one. Two corrections made during development are worth naming, because both were found by checking a computed result against a quantity whose exact value is known, and neither would have been visible in a residual history. The first is the wall-stress treatment of eq. (25): assembling the no-slip traction from the general face-gradient formula produced a normal component worth about two thirds of the Mach 2 laminar drag, where the exact value is zero. The second is the shock fix of eq. (20): on the Mach 2 aerofoil the unfixed HLLC flux carbuncled, putting the leading-edge pressure 28% above the Rayleigh–Pitot ceiling and giving $|C_L| = 4.6 \times 10^{-3}$ for a symmetric section at zero incidence. Both checks are now automated in `tools/sanity_checks.py`, which is the practical lesson: the useful tests are the ones that compare against an exact value, not against a previous run.

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